

AO and Remote Copy

Senthilkumar Muthusamy

Adaptive Optimization (AO)

- ▶ HPE 3PAR Adaptive Optimization (AO) enables the HPE 3PAR StoreServ to act as a hybrid storage array where the StoreServ can blend flash-based SSDs and HDDs to provide performance at an affordable price.
- ▶ HPE 3PAR Adaptive Optimization Software delivers the next generation in autonomic storage tiering by taking a fine-grained, highly automated approach to service-level improvement.



Brief overview of volume mapping

Chunklets

1 GB chunklets of physical disks treated like mini disks



SSD

FC

NL

Logical disks

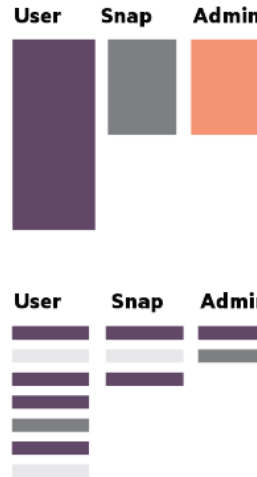
Chunklets from hundreds of drives form RAID sets



RAID 10, RAID 50 (1:2:8)
RAID 60 (4:2-14:2)

Volume spaces

User, snapshot, and admin spaces striped across logical disks regions



Virtual volumes

- Base or snapshot
- Full or thin provisioned
- Mapped to spaces



Virtual LUNs

Virtual volumes made visible to hosts as LUNs

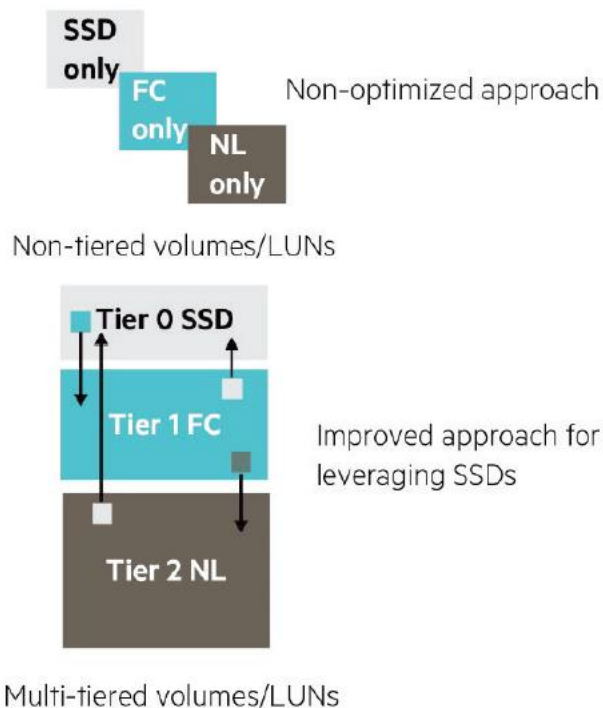


Non-tiered

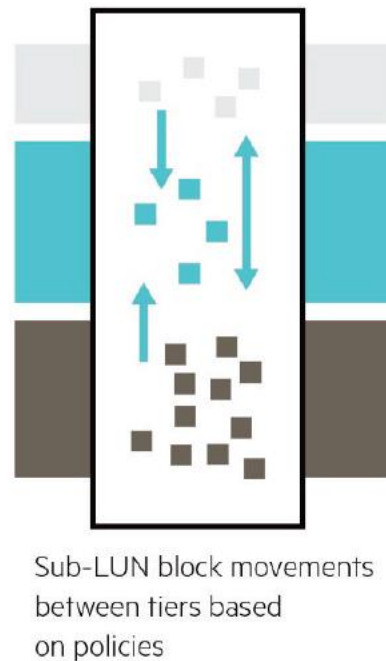
Tiered

Adaptively optimized volume

Block Level Tiering



HPE 3PAR Adaptive Optimization



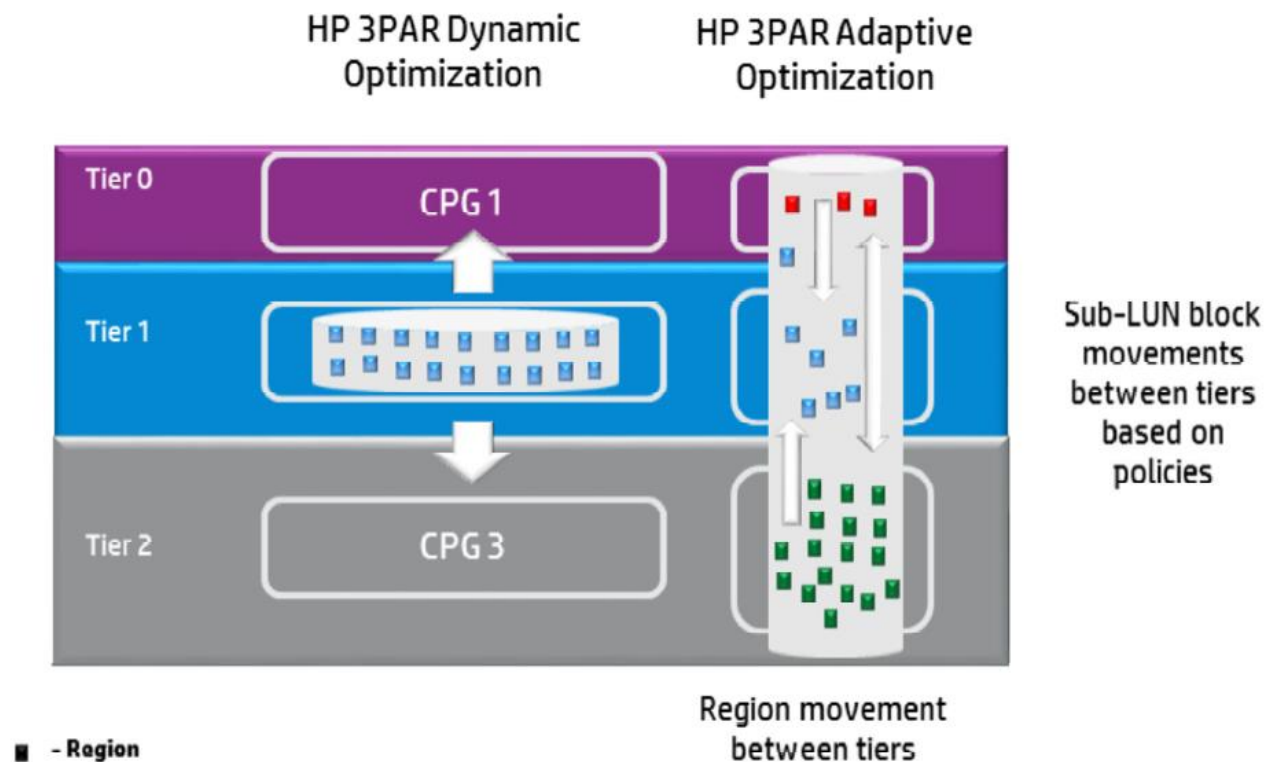
How AO Works ...

- ▶ HPE 3PAR Adaptive Optimization leverages data collected by HPE 3PAR System Reporter (SR) on node. The SR on node periodically collects detailed performance and space data that is used by AO for the following:
 - ▶ Analyze the data to determine the volume regions that should be moved between tiers
 - ▶ Instruct the array to move the regions from one CPG (tier) to another
 - ▶ Provide the user with reports that show the impact of Adaptive Optimization



Dynamic Optimization vs. Adaptive Optimization

Manual full versus automatic granular tiering



AO configuration

- Multiple AO configurations per array
 - Different configurations for different apps/domains
 - Different configurations can share same physical disks
 - Maximum 1024 configurations
- Up to three tiers per configuration
 - Each tier is a single CPG; volumes that use the CPGs are candidates for AO optimization
 - Tiers (drive type, RAID level, number of PDs, and so on) are defined by the CPG
 - Space limit is specified per tier in configuration (using CPG Growth Limit)
- Configuration includes execution schedules (cron-like) and measurement duration
- Choice of three modes:
 - Cost
 - Balanced (default)
 - Performance



Adaptive Optimization Specifications Matrix

Minimums/Maximums	Specifics
Max AO configurations	1024
Minimum tiers in AO config	2
Maximum tiers in AO config	3
Maximum capacity per AO config	Array capacity
Maximum data capacity movement per run	1 TiB OR 5% of AO config capacity
Minimum frequency for data sampling	30 minutes
Maximum IO density measurement	7 days
Minimum frequency for AO policy execution	1 hour



```

dxc-3par cli% showaocfg
-----CPG----- -Min(MB) - --Max(MB) -- -Warn(MB) -- -Limit(MB) -

Id      Name      T0      T1      T2 T0 T1  T2  T0 T1  T2  T0 T1  T2  T0 T1  T2
  Mode
  1 AO_CFG_DFLT SSD_r1 FC_r1 NL_r6  0  0   0 --- --- --- --- --- --- --- ---
Balanced
-----
-----
1      total

```

```

dxc-3par cli% createaocfg -t0cpvg SSD_r5 -t1cpvg FC_r6 -mode Performance testao1

```

```

dxc-3par cli%

```

```

dxc-3par cli% showaocfg
-----CPG----- -Min(MB) - --Max(MB) -- -Warn(MB) -- -Limit(MB) -

Id      Name      T0      T1      T2 T0 T1  T2  T0 T1  T2  T0 T1  T2  T0 T1  T2
  Mode
  1 AO_CFG_DFLT SSD_r1 FC_r1 NL_r6  0  0   0 --- --- --- --- --- --- --- ---
Balanced
  2 testao1 SSD_r5 FC_r6   ---  0  0 --- --- --- --- --- --- --- ---
Performance
-----
-----
2      total

```



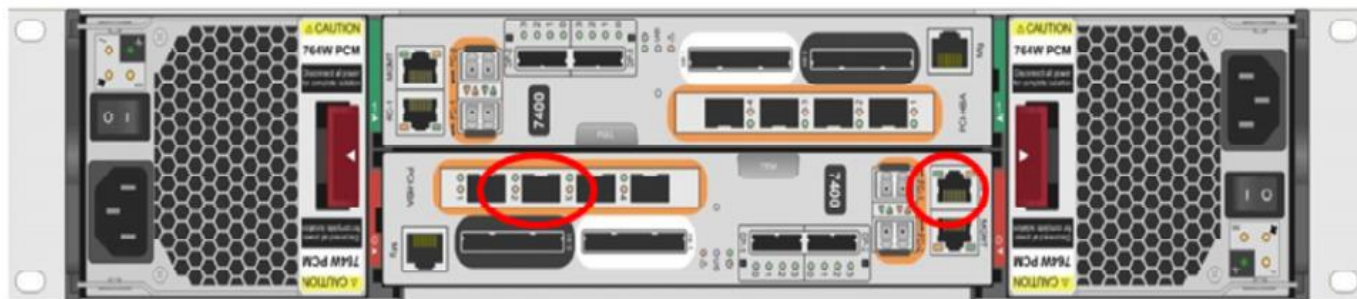

Remote Copy



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Remote Copy introduction

- Array-based remote replication solution for HP 3PAR Storage Array
- Simple setup and administration using CLI and Management Console (MC) and SSMC
- Limited Remote Copy functionality available in initial release of SSMC 2.0
- Synchronous, asynchronous periodic, and synchronous long distance (SLD) modes
- Remote Copy Fibre Channel (RCFC) and Remote Copy IP (RCIP) connectivity modes
- Fibre Channel over IP (FCIP) supported with additional hardware (for example, MPX200)
 - Asynchronous periodic replication only



7x00 series controller nodes shown

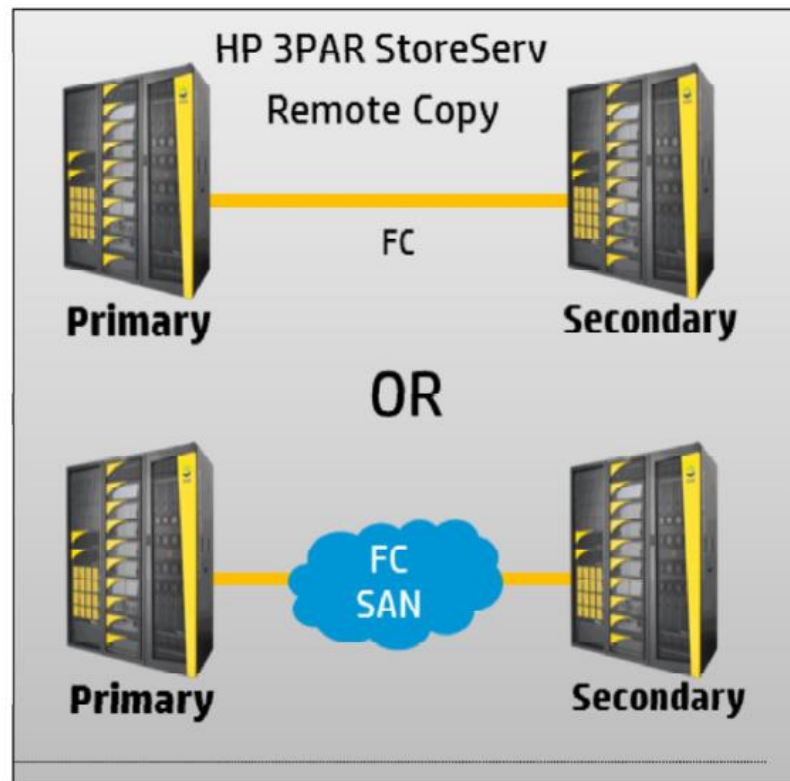
FC-based Remote Copy (RCFC)

High performance

- Used for campus-distance solutions
- Offers low latency and high bandwidth

Flexible

- Direct or FC SANs supported
 - Longwave links
 - Dense wavelength division multiplexing (DWDM)
 - FCIP bridging or routing



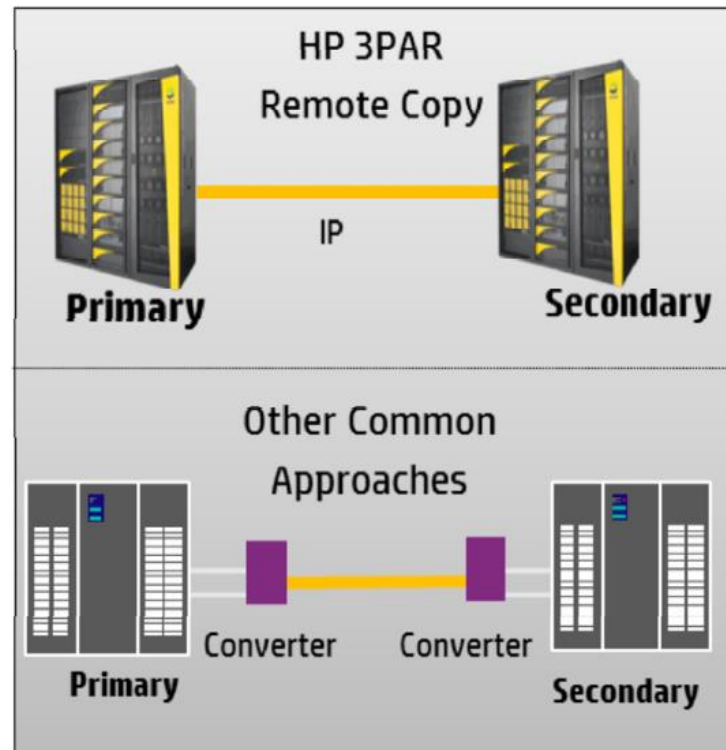
Native IP-based Remote Copy (RCIP)

Native IP eliminates need for expensive converters

- Distance flexibility
- Cost-effective
- No consulting services

Designed to be transport agnostic

- Native Gigabit Ethernet TCP/IP today
- Other protocols quickly assimilated



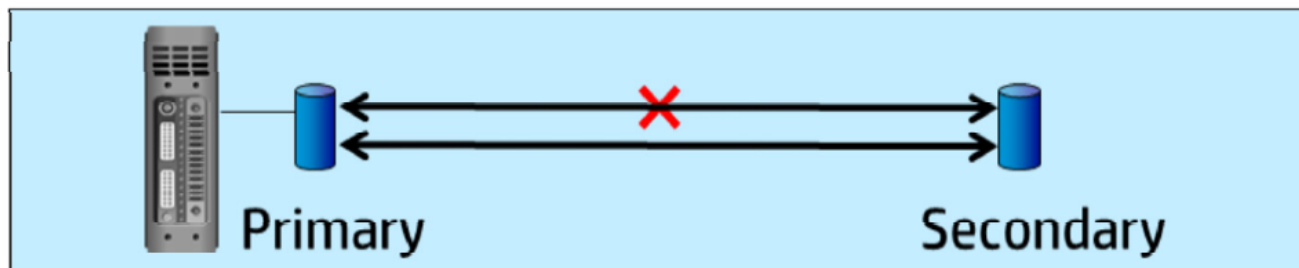
Active/Active replication links

Optimize link utilization, increase availability

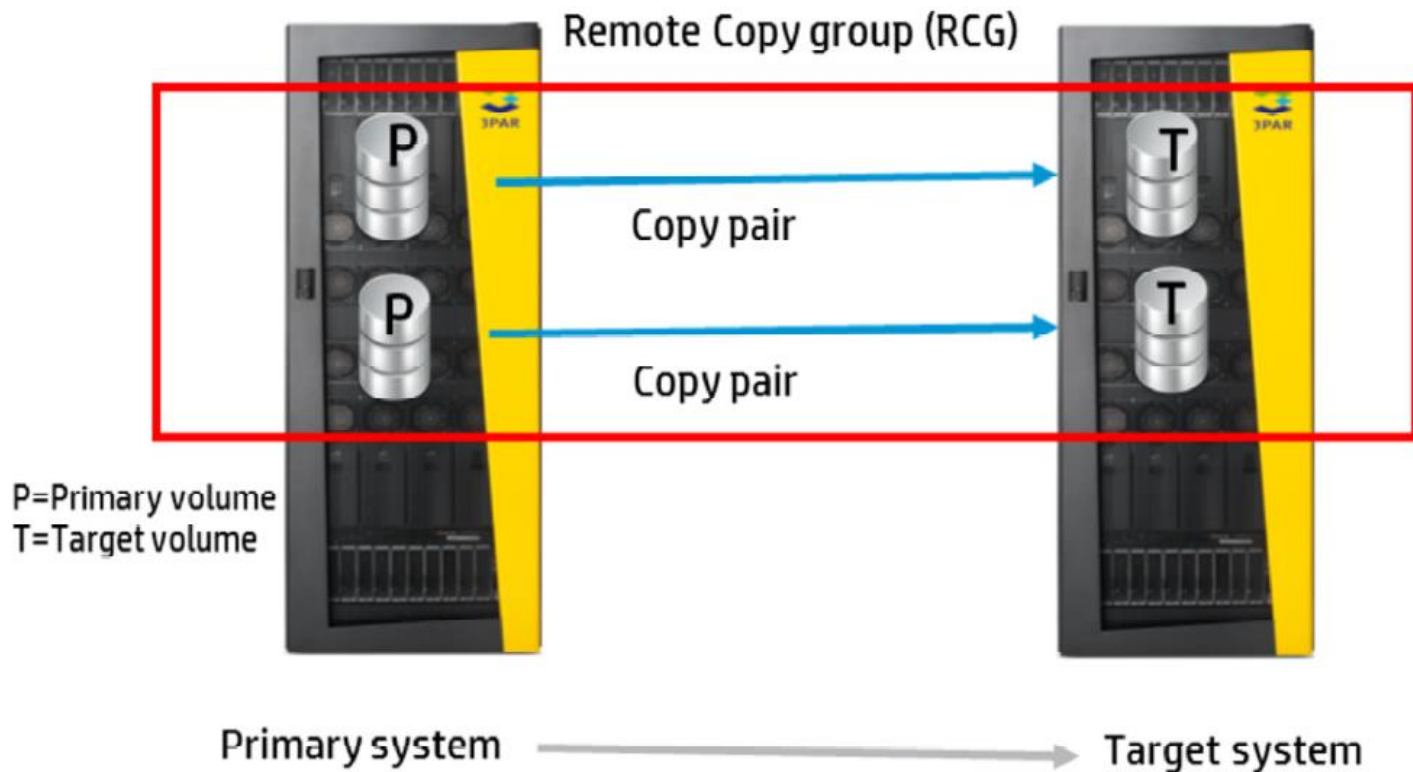
- Data transfers spread across all configured links
- Prevents bottlenecks
- Automatically rebalances load in case of link going down or restored
- Link Scalability

One RCIP link per node—Up to 8 per HP 3PAR array

Up to 4 RCFC links per node—Up to 32 per HP 3PAR array

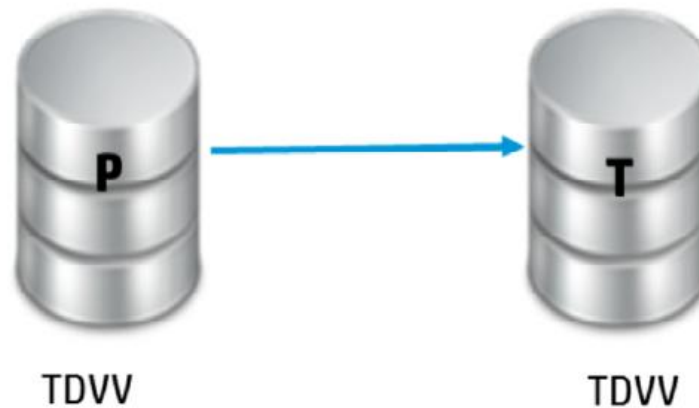


Remote Copy terminology



Thin Dedup aware

- Thin Dedup is supported as either source or target
- Using both *synchronous* and *periodic-async replication*

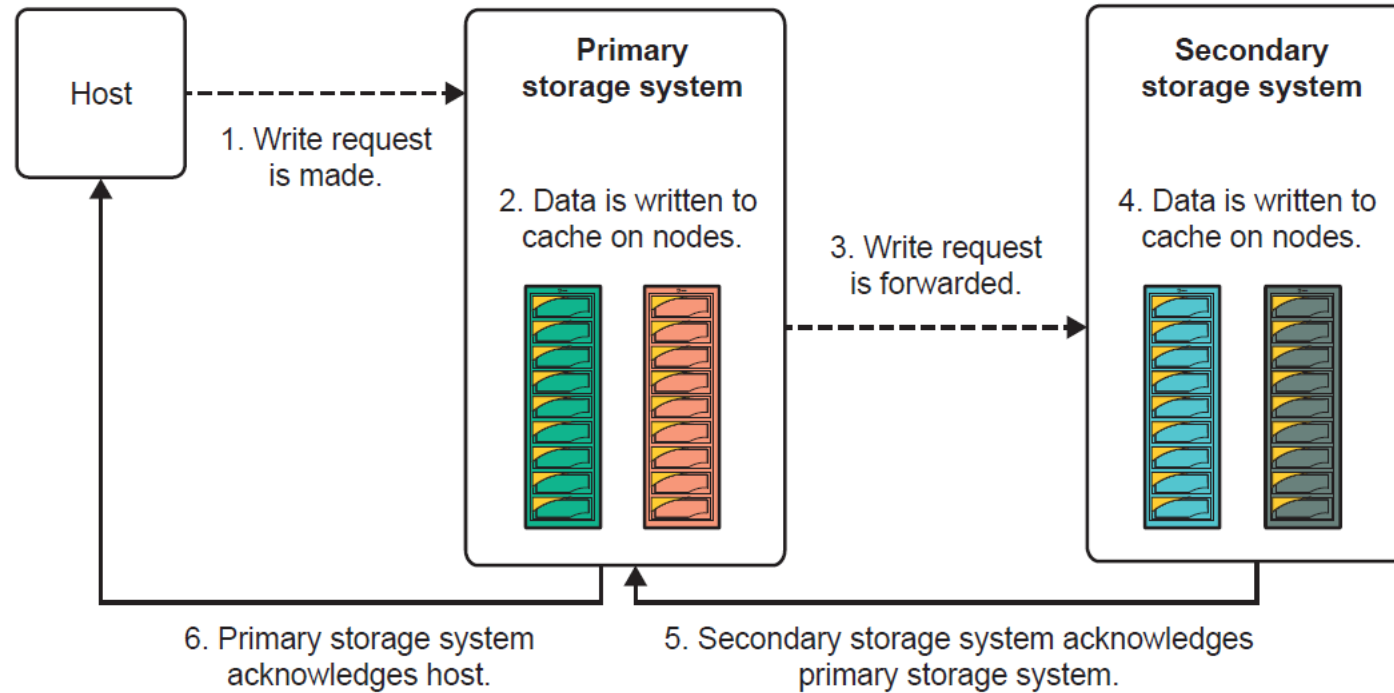


RC Types

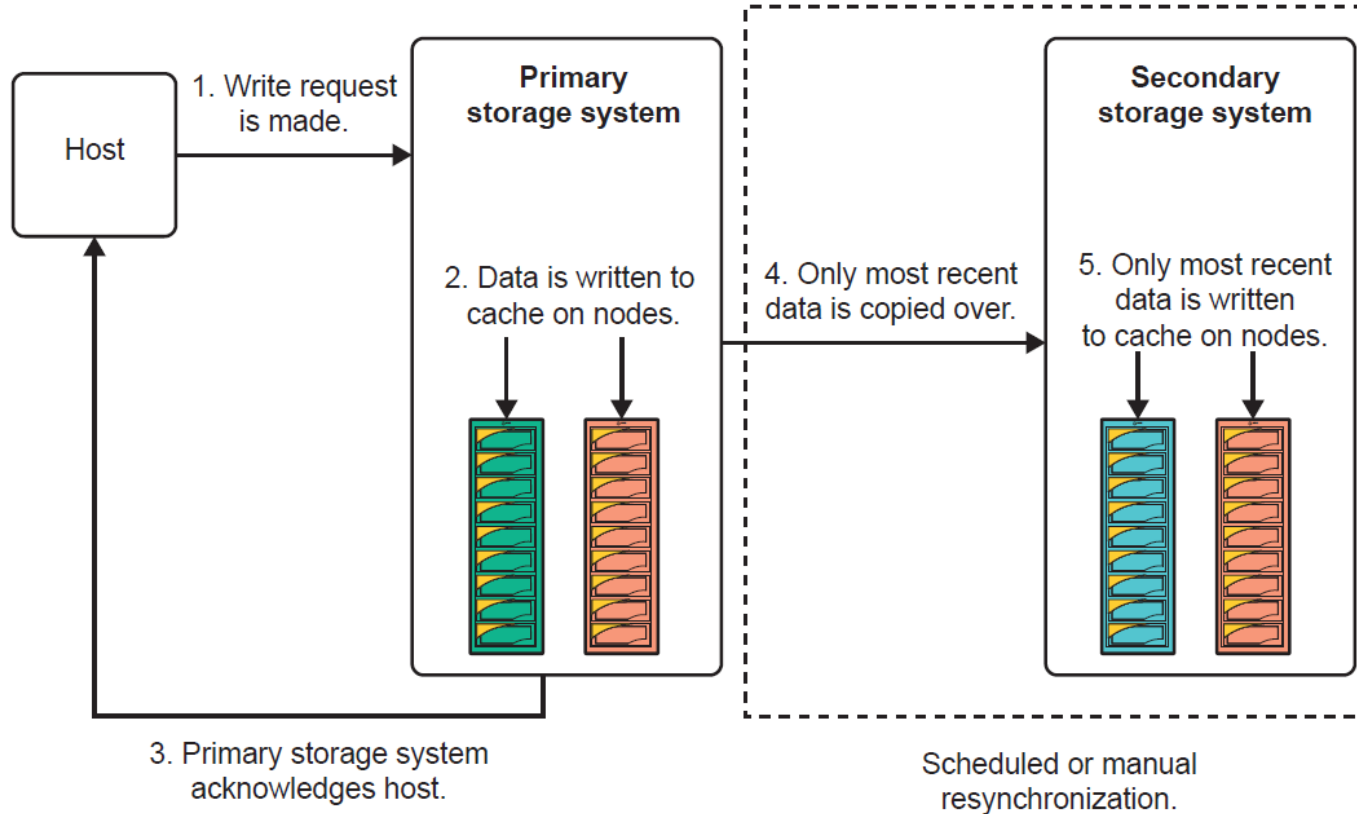
- ▶ **Synchronous Replication**
- ▶ **Periodic Asynchronous Replication**
- ▶ **Asynchronous Streaming Replication**
- ▶ **Synchronous Long Distance**



RC in Synchronous Mode



RC in Asynchronous Periodic Mode



Topology features

One-to-One (1:1)

Either in synchronous or asynchronous periodic mode or both

Unidirectional (Active/Passive) and bidirectional (Active/Active) support

Many-to-One (4:1)

Single array can be the target for multiple (up to 4) source arrays

One-to-Many (1:2)

One array can be primary to multiple (up to 2) target arrays

Different Remote Copy groups (you cannot send a given volume to two targets simultaneously)

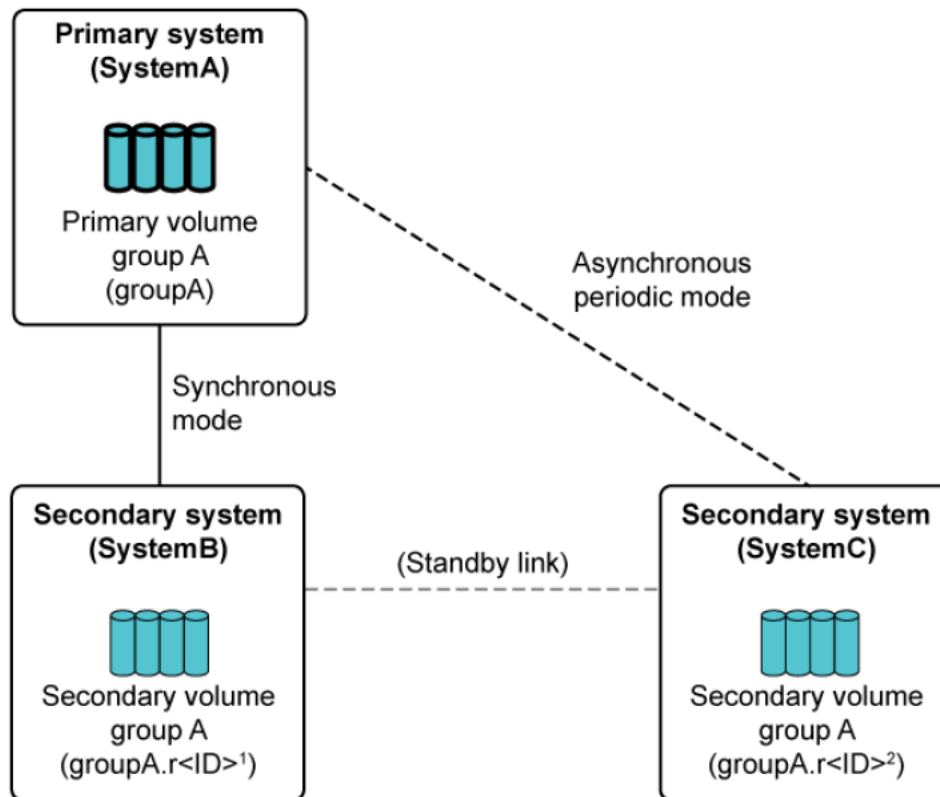
M-to-N (4x4)

Bidirectional data replication in a 4x4 fan-in, fan-out configuration

Synchronous long-distance Remote Copy configuration

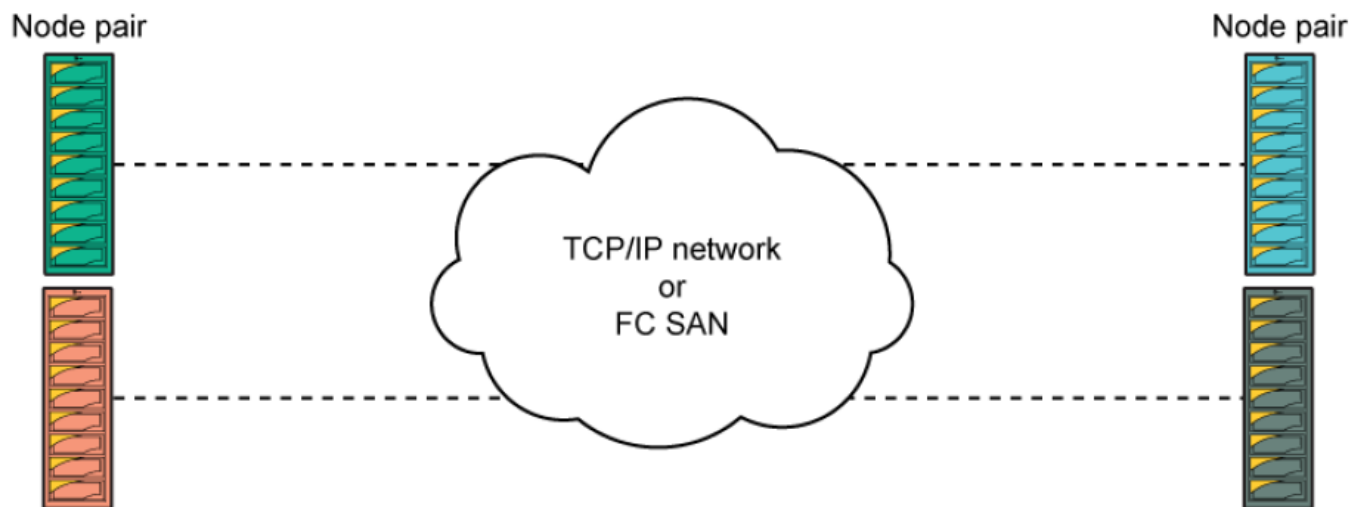
- ▶ A synchronous long distance (SLD) Remote Copy configuration is composed of three systems: two systems in a synchronous Remote Copy configuration and the two synchronous systems in an asynchronous periodic relationship with the third system.
- ▶ In an SLD configuration, Remote Copy groups from the primary system are replicated to the two separate target systems simultaneously.
- ▶ Data replicates to one target in synchronous mode and to the second in asynchronous periodic mode.
- ▶ The target systems must each have appropriate network connections to the source systems to support the replication mode being used.





Network transport methods for Remote Copy

Storage systems in a Remote Copy configuration are connected through a dedicated link or through a network.



HP 3PAR Remote Copy data integrity

Assured data integrity

Single volume

- All writes to the secondary volume are completed in the same order in which they are written on the primary volume

Multivolume group

- Volumes can be grouped together to maintain write ordering across sets of volumes
- Useful for databases or other applications that make dependent writes to more than one virtual volume
- Secondary groups and volumes are autonomically created or reconfigured and credentials inherited
- Maximum of 100 VVs

Autonomic replication groups

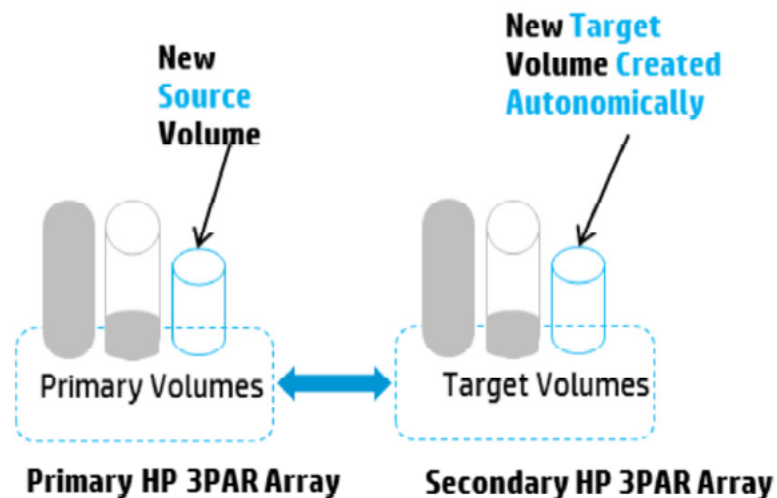
Greatly reduce operational cost and administrative time

Simplify, automate, and expedite storage replication configuration

- Configure replication group to match a CPG
- Replication policy applied automatically to new volumes
- Target volume on secondary storage created automatically

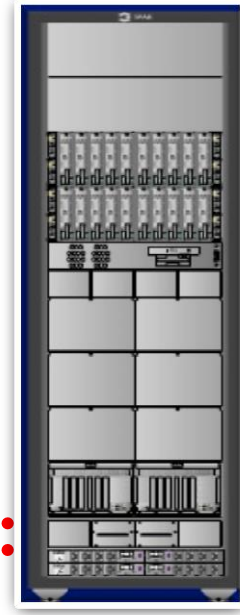
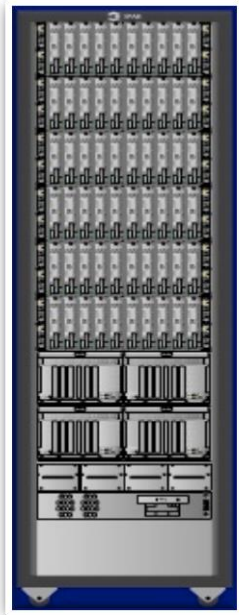
Increased agility

- Online increase in source volume size leads to automatic corresponding increase in target volume size



Remote Copy: Steps for Setup

1. StorServs can be any combination
2. Establish RC links
 - GigE
 - Fibre Channel
3. Start Remote Copy on both systems
4. Establish RC relationship on both systems
 - Each RC relationship is defined separately
5. Create RC group (synchronous or Periodic Asynchronous)
6. Add volumes (volumes must have snap space) to RC group
7. Start RC group (and synchronization)



Converting Standard Virtual Volumes

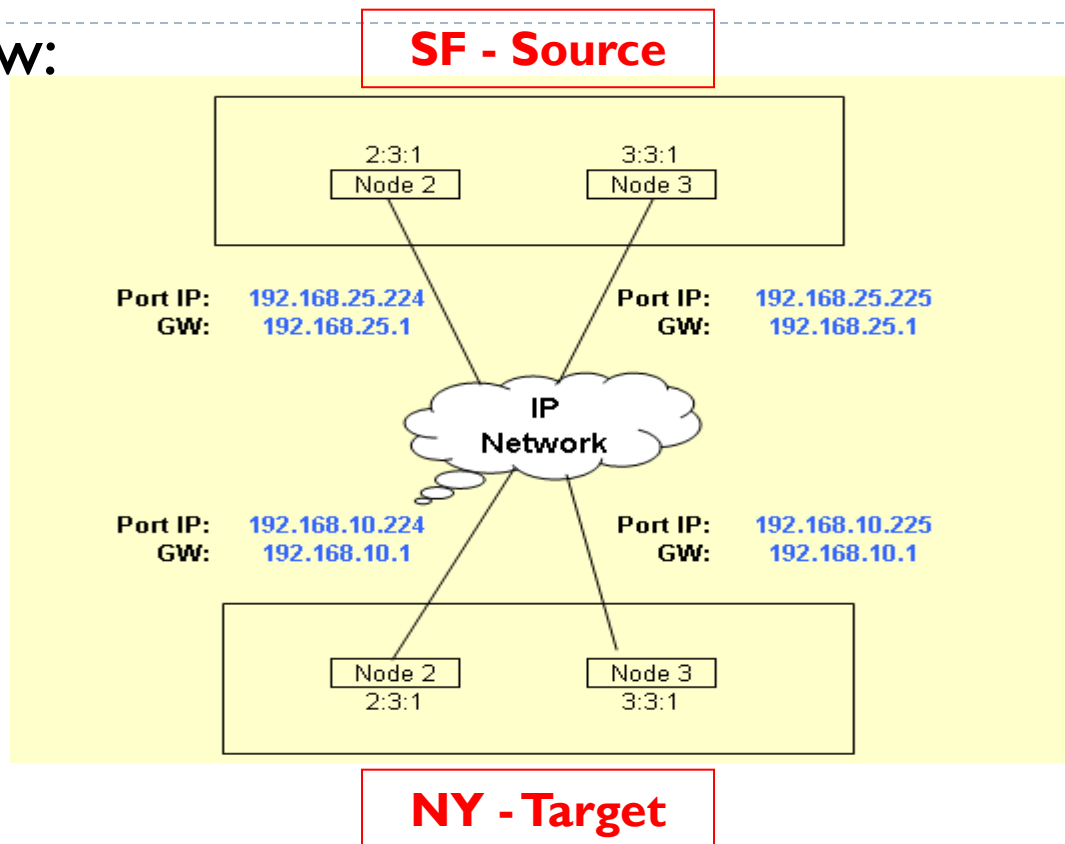
- ▶ Virtual Volumes that will be replicated must have snap space
- ▶ The Virtual Volume must exist on both StorServs and must be at least same size
- ▶ An often used naming convention for remote StorServ is .r



Initializing IP Setup

► Admin needs to know:

- Node:slot:port
- IP Addresses
- Gateway



IP Setup Primary (SF-Source)

```
SF-Source cli% controlport rcip addr 192.168.25.224  
255.255.255.0 2:3:1
```

```
SF-Source cli% controlport rcip addr 192.168.25.225  
255.255.255.0 3:3:1
```

```
SF-Source cli% controlport rcip gw 192.168.25.1  
2:3:1
```

```
SF-Source cli% controlport rcip gw 192.168.25.1  
3:3:1
```

```
SF-Source cli% controlport rcip mtu 9000 2:3:1
```

```
SF-Source cli% controlport rcip mtu 9000 3:3:1
```

```
SF-Source cli% controlport rcip speed auto 2:3:1
```

```
SF-Source cli% controlport rcip speed auto 3:3:1
```



IP Setup Secondary (NY-Target)

```
NY-Target cli% controlport rcip addr 192.168.10.224  
255.255.255.0 2:3:1
```

```
NY-Target cli% controlport rcip addr 192.168.10.225  
255.255.255.0 3:3:1
```

```
NY-Target cli% controlport rcip gw 192.168.10.1  
2:3:1
```

```
NY-Target cli% controlport rcip gw 192.168.10.1  
3:3:1
```

```
NY-Target cli% controlport rcip mtu 9000 2:3:1
```

```
NY-Target cli% controlport rcip mtu 9000 3:3:1
```

```
NY-Target cli% controlport rcip speed auto 2:3:1
```

```
NY-Target cli% controlport rcip speed auto 3:3:1
```



Verifying Connectivity For RC

```
mktg-ins2 cli% showport
N:S:P      Mode      State      Node_WWN      Port_WWN      Type
6:0:1 initiator ready 2FF70002AC000192 26010002AC000192 disk
6:0:2 initiator ready 2FF70002AC000192 26020002AC000192 disk
6:1:1 initiator ready 2FF70002AC000192 26110002AC000192 disk
6:1:2 initiator ready 2FF70002AC000192 26120002AC000192 disk
6:2:1 target loss_sync 2FF70002AC000192 26210002AC000192 free
6:2:2 target loss_sync 2FF70002AC000192 26220002AC000192 free
6:3:1 peer ready - 0002B3C787DE rcip
6:4:1 target ready 2FF70002AC000192 26410002AC000192 host
6:4:2 target loss_sync 2FF70002AC000192 26420002AC000192 free
7:0:1 initiator ready 2FF70002AC000192 27010002AC000192 disk
7:0:2 initiator ready 2FF70002AC000192 27020002AC000192 disk
7:1:1 initiator ready 2FF70002AC000192 27110002AC000192 disk
7:1:2 initiator ready 2FF70002AC000192 27120002AC000192 disk
7:2:1 target loss_sync 2FF70002AC000192 27210002AC000192 free
7:2:2 target loss_sync 2FF70002AC000192 27220002AC000192 free
7:3:1 peer ready - 0002B3C03E44 rcip
7:4:1 target ready 2FF70002AC000192 27410002AC000192 host
7:4:2 target loss_sync 2FF70002AC000192 27420002AC000192 free
mktg-ins2 cli%
```

Set up Remote Copy Targets and Links

```
SF-Source cli% startrcopy
```

```
SF-Source cli% creatercopytarget NY-Target IP  
2:192.168.10.224 3:192.168.10.225
```

```
SF-Source cli% showrcopy
```

```
NY-Target cli% startrcopy
```

```
NY-Target cli% creatercopytarget SF-Source IP  
2:192.168.25.224 3:192.168.25.225
```

```
NY-Target cli% showrcopy
```



Creating a Periodic Volume Group

```
SF-Source cli% creatercopygroup RCgroup1 NY-Target:periodic
```

Group name

Target Name—Created
with creatercopytarget
command

mode

```
SF-Source cli% setrcopygroup period 30m NY-Target RCgroup1
```

- ▶ Sets period for replication. In this command every 30 minutes to NY-Target for RCgroup1

Adding a VV to a Volume Group

```
SF-Source cli% admitrcopyvv Testdata RCgroup1 NY-Target:Testdata_r
```

Target Name—Created with
creatercopytarget command

Source VV
name

Group name

Target VV name

```
SF-Source cli% startrcopygroup RCgroup1
```

```
SF-Source cli% showrcopy
```



RC: Modifying RC Configuration

Adding or removing a volume to an RC group

► On SF-Source

```
SF-Source cli% stoprcopygroup group1
```

```
SF-Source cli% admitrcopyvv rctest3 group2  
NY-Target:rctest3.r
```

```
SF-Source cli% startrcopygroup group1
```

1. Stop the RC Group
2. Admit the rcopyvv(s)
3. Start the RC group





Dynamic Optimization



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HPE 3PAR Dynamic Optimization

- ▶ HPE 3PAR Dynamic Optimization is an optional feature that allows you to improve the performance of virtual volumes without interrupting access.
- ▶ Use this feature to avoid over-provisioning for peak system usage by optimizing the layout of your virtual volumes.
- ▶ With dynamic optimization, you can change the parameters, RAID levels, and set sizes of the virtual volume by associating the virtual volume with a new CPG.
- ▶ For example, when a system is upgraded by adding nodes, cages, or physical disks, the initial volume and LD layouts may no longer be optimal for the new system configuration.
- ▶ Updating the system layout optimizes the use of all physical resources in the system at a given time.



-
- ▶ Dynamic optimization may improve system performance in several ways:
 - ▶ **Volume layout changes after hardware upgrades**
 - ▶ **Volume RAID level changes**
 - ▶ **Volume fault-tolerance changes**
 - ▶ **CPG and volume growth configuration changes**



DO Phases

- ▶ The dynamic optimization automated tuning process has three phases:
- ▶ **1. Internode tuning phase**—Analyze the system and detect virtual volumes that are not correctly balanced between nodes. If virtual volumes are not balanced correctly, the volumes are tuned to correct the imbalance.
- ▶ **2. Intranode tuning phase**—Analyze the system and detect any chunklet imbalance between physical disks associated with the same node. After the analysis, chunklets are moved from over-used physical disks to under-used physical discs associated with the same node.
- ▶ **3. CPG analysis phase**—Analyze the system and verify that logical disks associated with a CPG have the same characteristics as the CPG. If the LD characteristics do not match the CPG, the LD is modified to match the CPG characteristics.



3PAR Dynamic Optimization Benefit For Storage Administration

- ▶ Allows for a non-disruptive re-layout of Virtual Volumes on the 3PAR StorServ (move volumes to different physical disks to accommodate new user demands)
- ▶ Promotes system optimization through improved utilization of all physical resources for the current configuration (Take advantage of a H/W upgrade)
- ▶ Allows for altering of “service levels” associated to a Virtual Volume (change RAID levels, set sizes, spare chunklets)



How to Use Dynamic Optimization (I of 4)

3PAR InForm Management Console

File View Actions Window Help

Connect... Refresh Maximize Home Back Forward Export Data... Virtual Volume Virtual Copy Physical Copy Export... Edit... Remove...

Provisioning

- VMWare
 - CPGs
 - Virtual Volumes
 - Virtual Volume Sets
 - VLUNs

Common Actions

- Create Virtual Volume...
- Create Virtual Copy...
- Create Physical Copy...
- Create CPG...
- Create VLUN...

Systems

- Hosts
- Provisioning
- Layout Grid
- Remote Copy
- Performance

Connected to ob

Provisioning : Domains : VMWare : Virtual Volumes

Summary Virtual Volumes Alerts

Summary 4 rows Filter Clear <Any column> contains: Clear

Name	Domain	State	Type	Provisioning	RAID	Virtual Size (GiB)	Used User Size (% Virtual)	Reser Copy
ESX_Disk_1_NL	VMWare	Normal	Basic	Thin	RAID 5	500.000	80%	
ESX_Disk_2_FC_10K	VMWare	Normal	Basic	Thin	RAID 5	500.000	0%	
ESX_Disk_3_FC_15K	VMWare	Normal	Basic	Thin	RAID 5	500.000	1%	
Test	VMWare	Normal	Basic	Thin	RAID 5	100.000	0%	
						1,600.000		

Virtual Volume Detail:

Summary Settings VLL

General

New Alerts Recent Tasks

System	Severity	Time	Message
3par2	Info	77 Oct 12, 2010 13:19:12 CDT	Thin provisioned VV ESX_Disk_1_NL has reached allocation warning
3par2	Degraded	79 Oct 02, 2010 23:36:03 CDT	Port 3:1:2 Degraded (Target Mode Port Went Offline)
3par2	Degraded	78 Oct 02, 2010 23:36:03 CDT	Port 3:1:2 Degraded (Target Mode Port Went Offline)

Message

How to Use Dynamic Optimization (2 of 4)

Tune Virtual Volume(s) : 3par2 (1302031) : ESX_Disk_1_NL

Steps

1. General
2. Summary

General

General

System: 3par2 (1302031)

Domain: VMWare

Allocation Space: User Space (Fully Provisioned from CPG)

Virtual Volume(s):

- ESX_Disk_1_NL
- ESX_Disk_2_FC_10K
- ESX_Disk_3_FC_15K
- Test

1 Virtual Volume to be tuned: ESX_Disk_1_NL

CPG

Current: ESX_Cluster_R5_NL
(RAID 5, NL, 4.125 GiB Used, 51.000 GiB Limit)

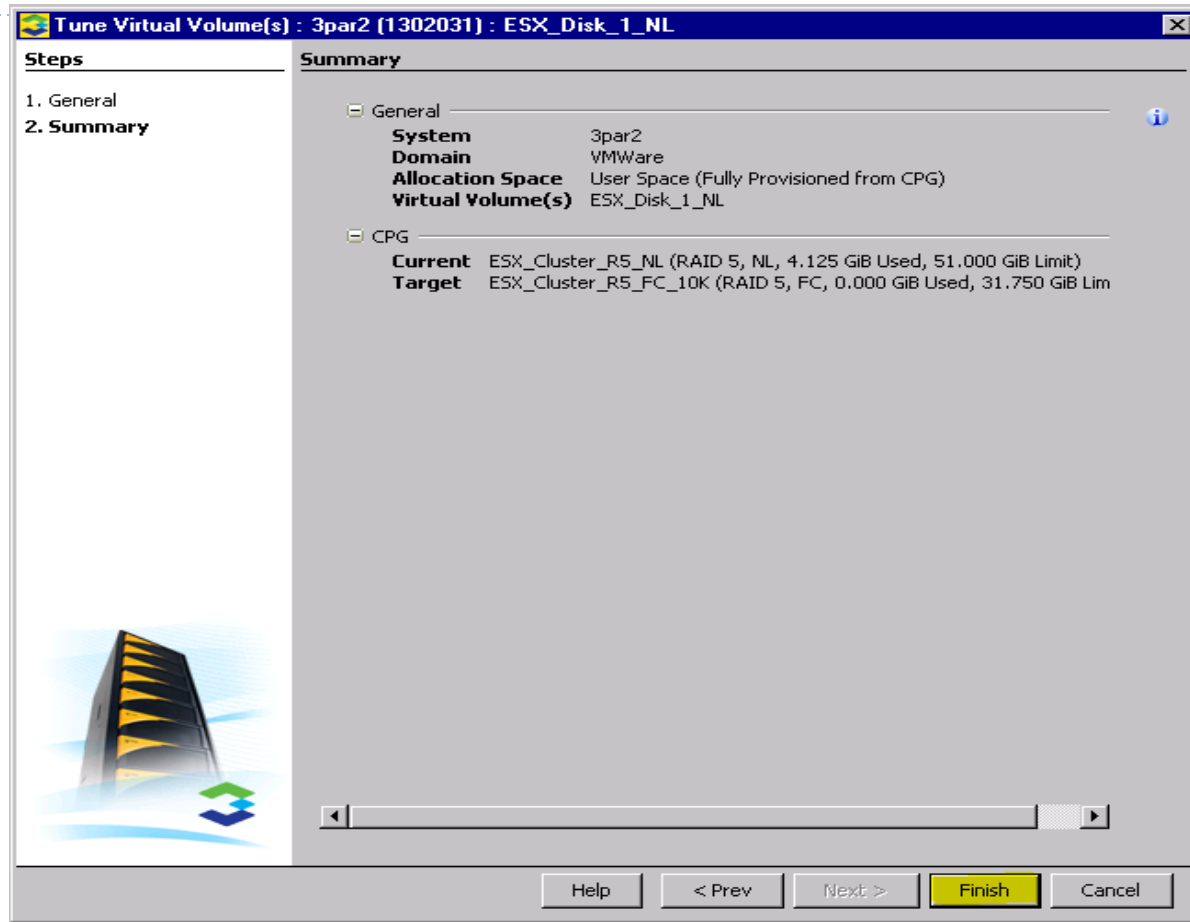
Target:

- ESX_Cluster_R5_FC_10K (RAID 5, FC, 0.000 GiB Used, 31.750 GiB Limit)
- ESX_Cluster_R5_FC_15K (RAID 5, FC, 0.000 GiB Used, 24.250 GiB Limit)

Target is required

Help < Prev Next > Finish Cancel

How to Use Dynamic Optimization (3 of 4)



How to Use Dynamic Optimization (4 of 4)

The screenshot displays the VMware vSphere Client interface. On the left is a navigation pane with the following menu items: Systems, Hosts, Provisioning (highlighted), Layout Grid, Remote Copy, and Performance. Below these are icons for a checkmark, settings, a lock, and a list. The main window is titled "Virtual Volume Details" and contains tabs for Summary, Settings, VLUNs, Layout, and Alerts. The "General" tab is active, showing a "Capacity" bar. Below this is a sub-tabbed interface with "New Alerts", "Recent Tasks" (highlighted), and "My Connections". The "Recent Tasks" tab displays a table with the following data:

System	ID	Type	Name	Status	Progress	Start Time	Finish Time	Dur
3par2	3966	Tune Copy Space	ESX_Disk_1_NL	Running	0%	Oct 18, 2010 12:13:30 CDT	--	

At the bottom of the window, a status bar shows the message: "Started Tune Virtual Volume ESX_Disk_1_NL on System 3par2".

Completion of Dynamic Optimization

Provisioning : Domains : VMWare : Virtual Volumes

Summary Virtual Volumes Alerts

Summary 4 rows Filter Clear

<Any column> contains: Clear

Name	Domain	State	Type	Provisioning	RAID	Virtual Size (GiB)	Used User Size (% Virtual)	Reserved Copy Size (GiB)	Used Copy Size (% Virtual)
ESX_Disk_1_NL	VMWare	Normal	Base	TPVV	RAID 5	500.000	80%	4.125	0%
ESX_Disk_2_FC_10K	VMWare	Normal	Base	TPVV	RAID 5	500.000	0%	--	--
ESX_Disk_3_FC_15K	VMWare	Normal	Base	TPVV	RAID 5	500.000	1%	--	--
Test	VMWare	Normal	Base	TPVV	RAID 5	100.000	0%	0.000	0%

1,600.000 4.125

Virtual Volume Details

Summary Settings VLUNs Layout Alerts

User Space Allocation

Device Type	FC
Device Speed	10 K
RAID Type	RAID 5
Configured Availability	Cage
Availability	Cage
Set Size	4
Step Size	128 KiB

Copy Space Allocation

Device Type	FC
Device Speed	0 K
RAID Type	RAID 0
Configured Availability	Chunklet
Availability	Port
Set Size	1
Step Size	256 KiB

New Alerts Recent Tasks My Connections

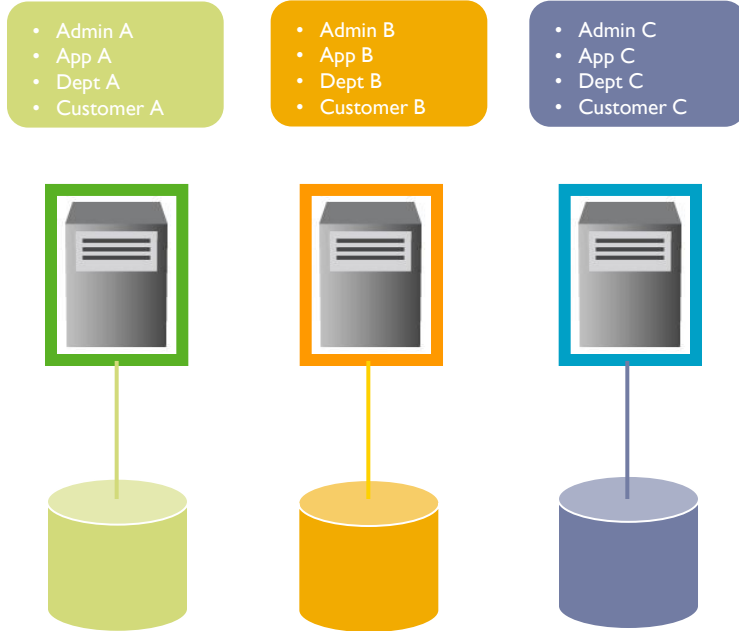
System	ID	Type	Name	Status	Progress	Start Time	Finish Time	Duration
3par2	3966	Tune Copy Space	ESX_Disk_1_NL	Completed	--	Oct 18, 2010 12:13:30 CDT	Oct 18, ...	48 minutes



Virtual Domain

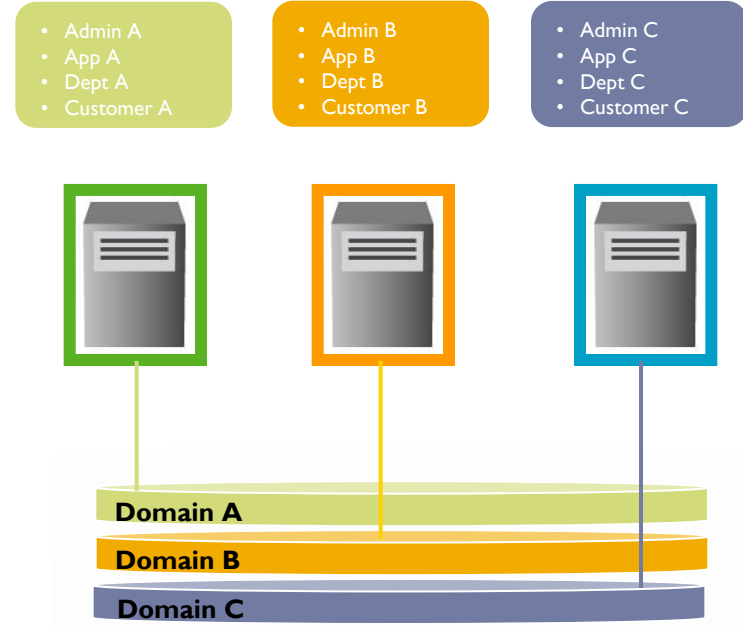
What are 3PAR Virtual Domains?

Multi-Tenancy with Traditional Storage



Separate, Physically-Secured Storage

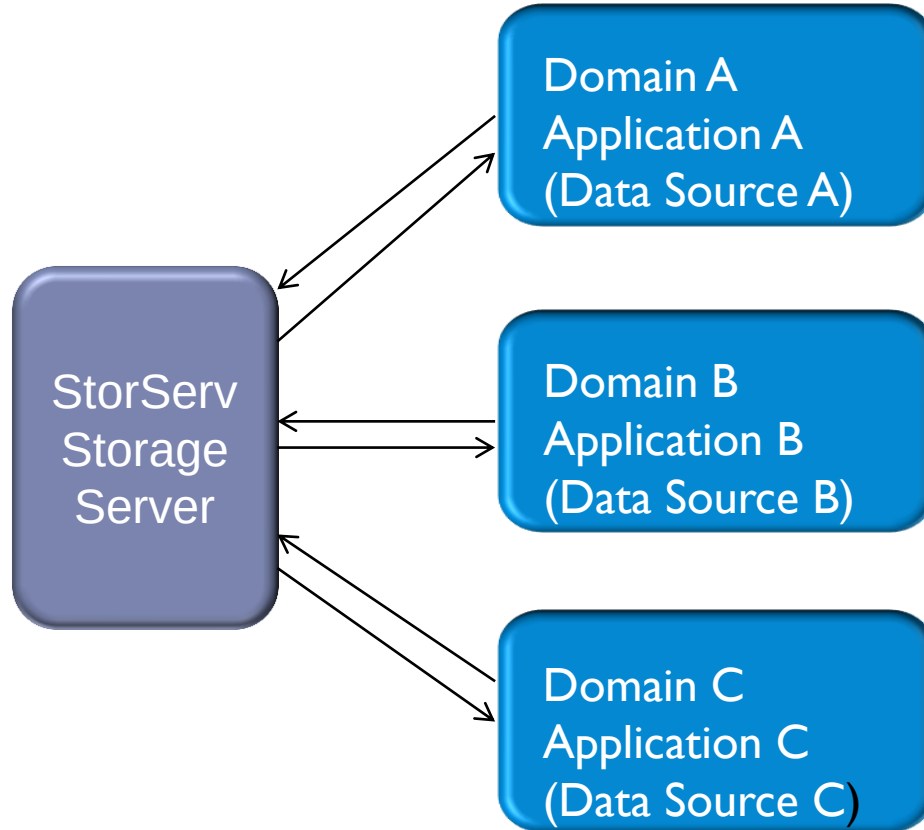
Multi-Tenancy with 3PAR Domains



Shared, Logically-Secured Storage

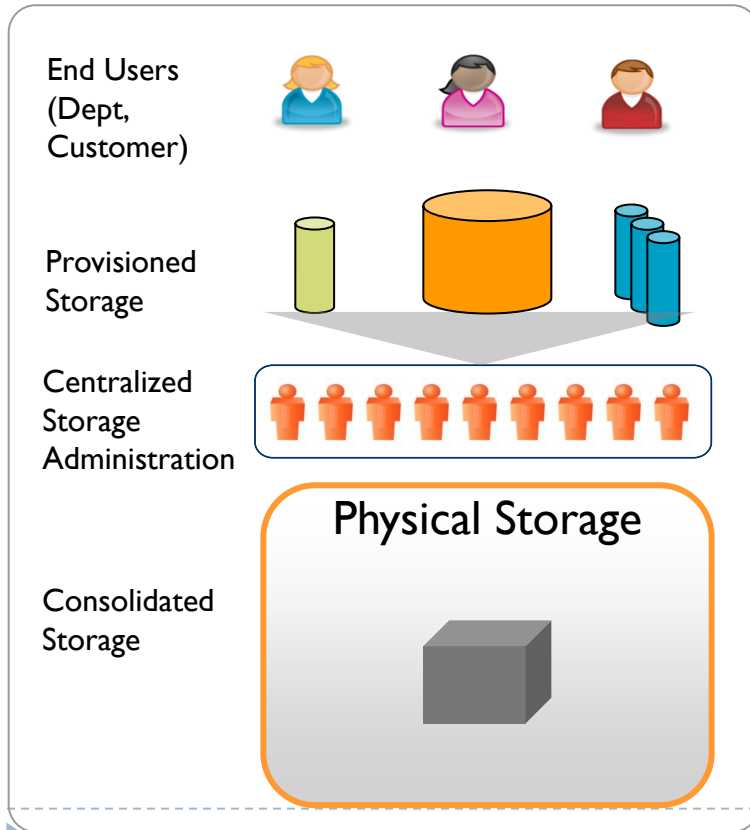
Virtual Domains Overview

- Allows fine grained access control on an Storage Server
- Up to 1024 domains within an StorServ Storage Server
- Each Domain can be dedicated to a specific application
- A sub set of the StorServ Storage Server users have varying privileges over the domains
- Requires an HP 3PAR Virtual Domain License

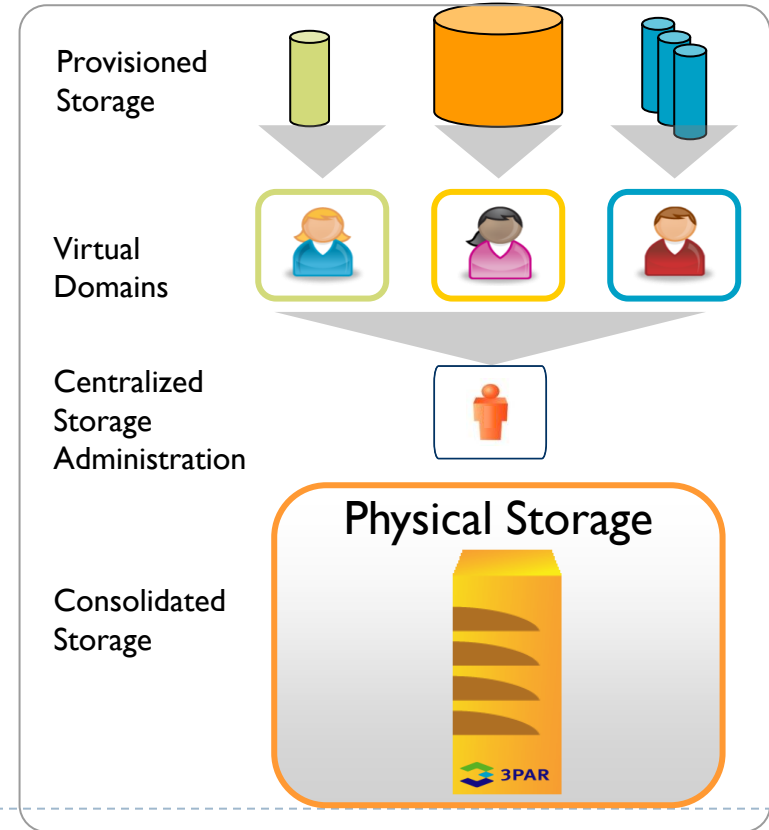


When Do You Sell Virtual Domains?

Centralized Storage Admin with Traditional Storage



Self-Service Storage Admin with 3PAR Virtual Domains



3PAR Domain Objects

Basic Domain Objects

- Common Provisioning Groups (CPGs)
- Hosts
- Remote Copy groups.

▶ Derived Domain Objects

- Virtual Volumes (VV)
- VV Sets
- VLUNs
- Hosts (Including Paths)
- Host Sets

- ▶ Objects are domain specific so users cannot export VVs to a host outside of their assigned domain
-

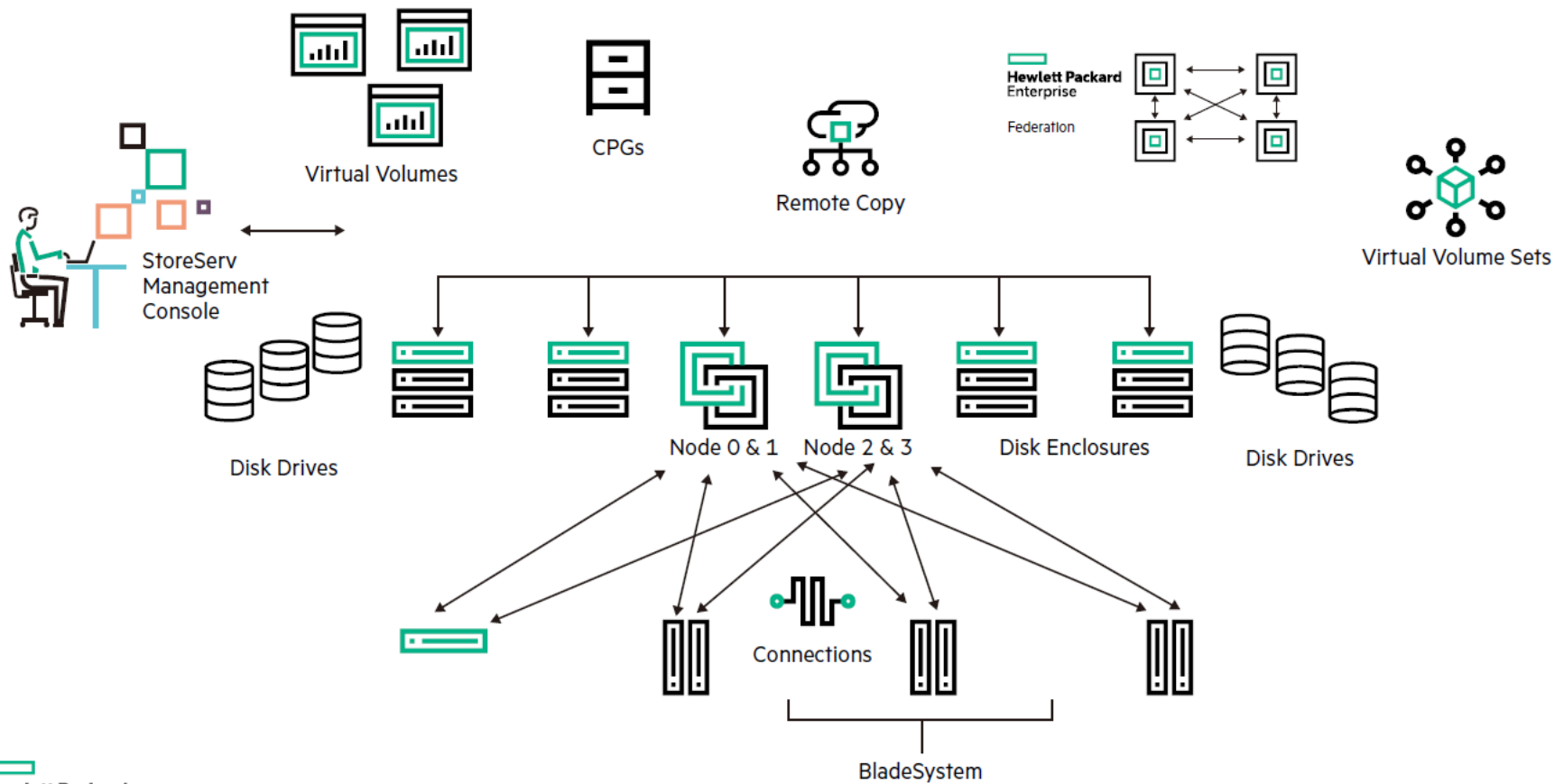




HPE 3par System Reporter



Senthilkumar Muthusamy



System Reporter

- ▶ System Reporter is optional software that samples the space and performance data, storing it in a database and providing historical capacity and performance reporting for collection of HP 3PAR StoreServ Storage Systems.
- ▶ These reports can be used for several purposes including:
 - ▶ Performance Analysis
 - ▶ Capacity Planning
 - ▶ Active Monitoring
 - ▶ Scheduled reports



System Reporter



3PAR StoreServ



GENERAL

Dashboard

Activity

Schedules

Settings

BLOCK PERSONA

Hosts

Host Sets

Virtual Volumes

Virtual Volume Sets

Common Provisioning Groups

FILE PERSONA

File Shares

File Stores

Virtual File Servers

File Provisioning Groups

Persona Configuration

STORAGE OPTIMIZATION

Adaptive Optimization

REPLICATION

Remote Copy Configurations

Remote Copy Groups

STORAGE SYSTEMS

Systems

SYSTEM REPORTER

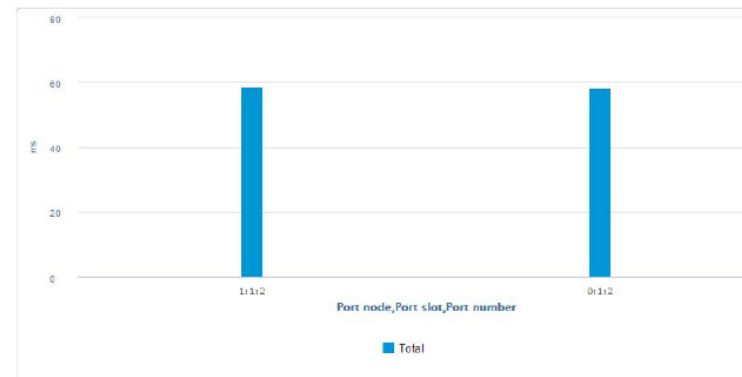
Reports

SECURITY

Users

AT Time Report - Host Port Performance

Service Time: 20-Apr-2015 00:00:00 o'clock PDT



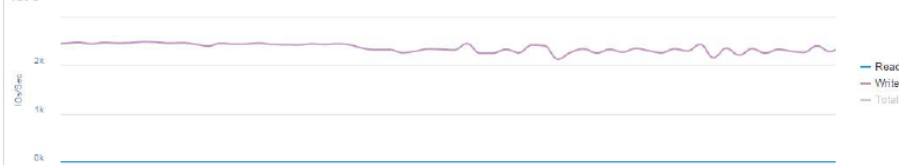
Host Port Performance

Charts

Actions

System: **mtkg-0062** Report type: Sampling Time interval: Hi-res Metric types: IOPs, Service time, Bandwidth
Component: Ports

IOPs



Service Time

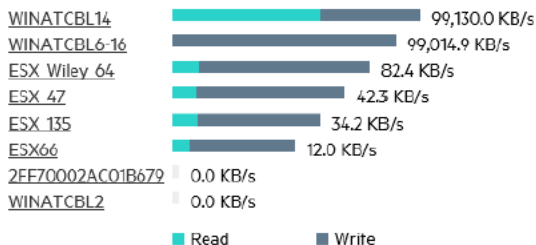


- Performance Templates
- Bird-View with drill-down from daily to hi-res
- One-click at-time reports



Dashboard All systems ⚙️

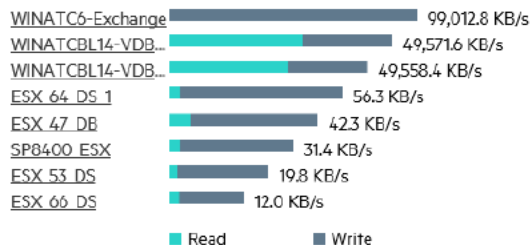
Daily Top Host Bandwidth



Storage Systems 3 >



Daily Top Volume Bandwidth



Performance >

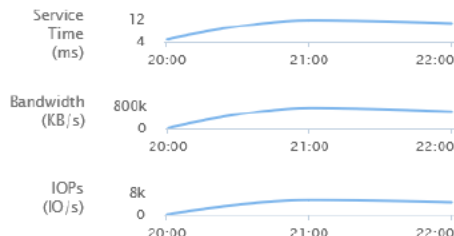


System >

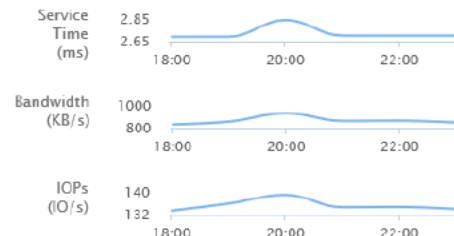
ATCHOU8400

Model HPE_3PAR 8400
 3PAR serial number 2M263305NL
 HPE 3PAR OS version 3.3.1.215 (GA)+P01,P02,P04
 State Normal
 Details none

System Performance 8400 >



System Performance 7200 >



Capacity Efficiency >

Common Actions And Views

Total Capacity >



Reports 2 matches

Component

☐ All☐ Controller nodes☐ Ports☒ Exported volumes☐ Physical drives☐ Priority optimization☐ Virtual volumes☐ Common provisioning groups☐ Remote copy☐ Adaptive optimization☐ SMB protocol☐ NFS protocol☐ File persona networking☐ File provisioning group☐ File persona CPU☐ File persona memory☐ System capacity

+ Create report

● Name

● Exported Volumes Compare by Performance

○ Real Time Exported Volumes Performance

Type

System

Data Type

All

Reset

ATCHOU8400

Exported Volumes Compare by Performance

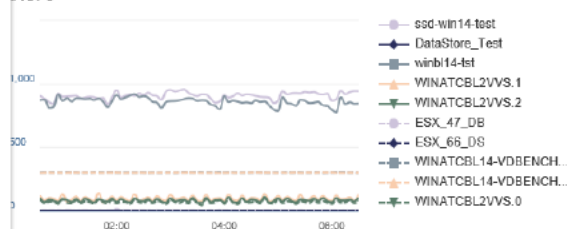
Charts

Actions

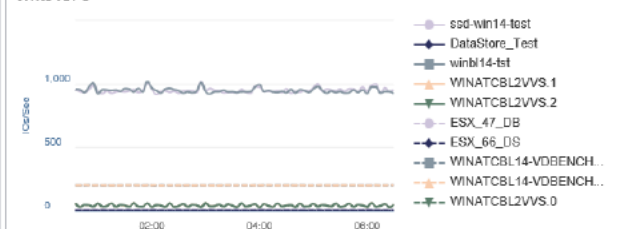
IOPS, Service time, Bandwidth...



IOPS



Write IOPS

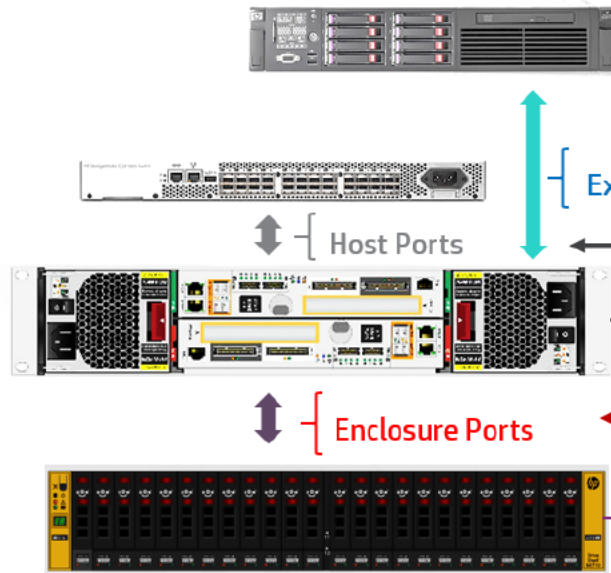


IOPS



Read Service Time





Name	Data Type	Category
Controller Node Cache - Performance Statistics	Historical	Performance
Controller Node CPU - Performance Statistics	Historical	Performance
Cumulative IO Density - AOCFG	Historical	Performance
Cumulative IO Density - CPG	Historical	Performance
Enclosure Port - Performance Statistics	Historical	Performance
Exported Volumes - Performance Statistics	Historical	Performance
File Persona Networking - Performance Statistics	Historical	Performance
FPG - Performance Statistics	Historical	Performance
Host Port - Performance Statistics	Historical	Performance
IO Density - AOCFG	Historical	Performance
IO Density - CPG	Historical	Performance
NFS Protocol - Performance Statistics	Historical	Performance
Physical Drive - Performance Statistics	Historical	Performance
Remote Copy Links - Performance Statistics	Historical	Performance

.srdata maintenance

System Reporter retains all collected data within the HPE 3PAR StoreServ array. A volume is created upon array initialization and identified as .srdata. The .srdata database will vary in size dependent upon the size of the array; by default the following database sizes will be created upon array initialization:

SR database initial created size

Table 1. Node to database size

Number of CPU nodes	Size of .srdata database
2	60 GB
4 or 6	80 GB
8i	100 GB

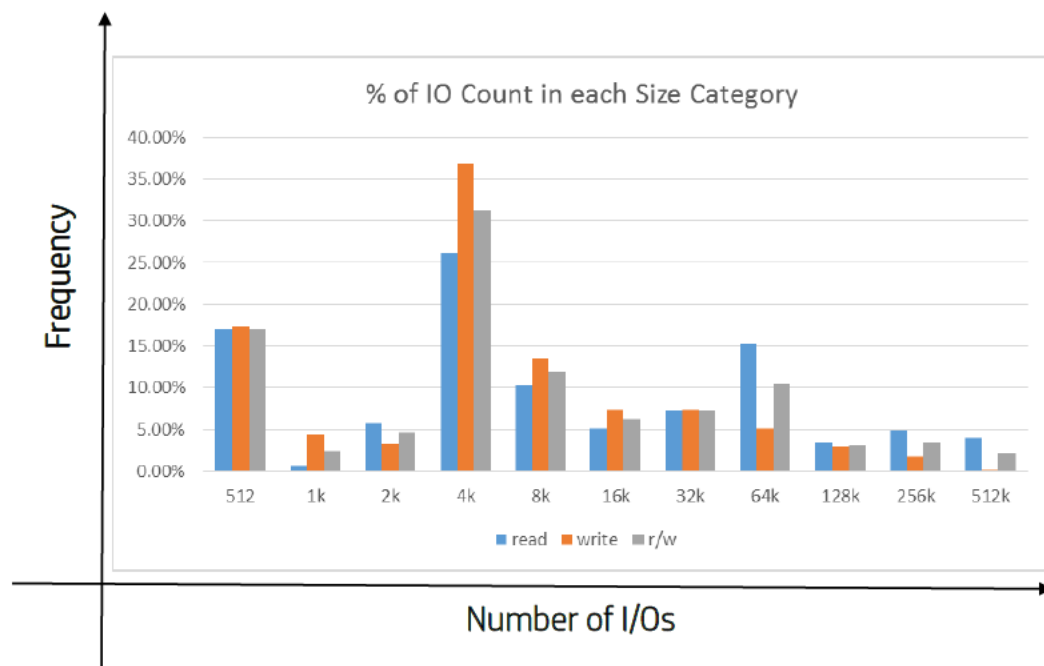


	Real-Time CLI		Historical CLI		SSMC
Front-End Performance					
Host Ports	statport -host	histport -host	srstatport	srhistport	Host Ports
VLUNs	statvlun	histvlun	srstatvlun	srhistvlun	Exported Volumes
FCoE Protocol	statfcoe		NA		Host Ports (fcoe)
iSCSI Protocol	statiscsi		NA		Host Ports (iSCSI)
iSCSI Session	statiscsession		NA		NA
FC RC Links	statport -rcfc	histport -rcfc	srstatrcopy		Remote Copy
Controllers Performance					
Cache	statcmp		srstatcmp	NA	Controller Node Cache
CPUs	statcpu		srstatcpu	NA	Controller Node CPU
LDs	statld	histld	srstatld	srhistld	NA
VVs	statvv	histvv	NA		NA
RC Groups	statrcopy		srstatrcvv		Remote Copy
RC VVs	statrcvv	histrcvv	srstatrcvv		Remote Copy
Back-End Performance					
Disk Ports	statport -disk	histport -disk	srstatport	srhistport	Enclosure Ports
Physical Disks	statpd	histpd	srstatpd	srhistpd	Physical Drives



Using histogram data

An **histogram** is a graphical representation of the distribution of numerical data. 3PAR records IO time and IO size of every single IO processed for this very reason histogram data can be very powerful to show your customer how well a system is behaving or what hosts/volumes have poor service times. The best way of reading the data is looking at percentages.



The graph shows the amount of I/O request of a given page size that the array has served

Performance Isolation Examples

	Exported Volumes	Host Ports	Physical Drives	Action
Service Times	Low	Low	Low	None required
Service Times	High	Low	Low	Verify host system resources and host hba settings
Service Times	High	Low	High	Verify Controller Cache stats
Service Times	High	High	Low	Verify connectivity between host and array (possible port switch problem /ISL congestion)



System Capacity

- This report is a good way to check the system or tier overall historical allocation and space distribution.
- When multiple tiers are present it makes sense to create a System Report for any given tier.

✓ System Capacity - SSDs

Charts ▾

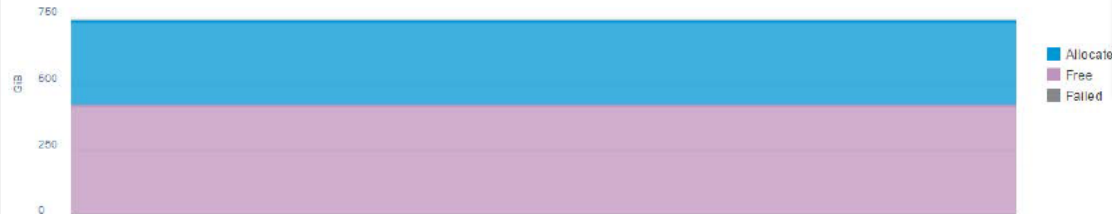
Action

■ Update report Completed

3paradm 26-Apr-2015 18:39:05 o'clock PDT

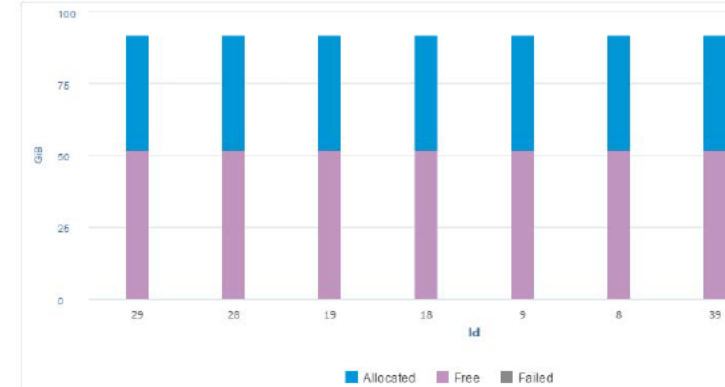
System	Report type	Time interval	Metric types	Space utilization
mktg-eos62	Sampling	Daily		
Component	Physical drives			

System Capacity



AT Time Report - System Capacity - SSDs

System Capacity: 26-Apr-2015 00:00:00 o'clock PDT



In SSMC 2.2 the free and allocated order will be inverted

Physical Drives Capacity

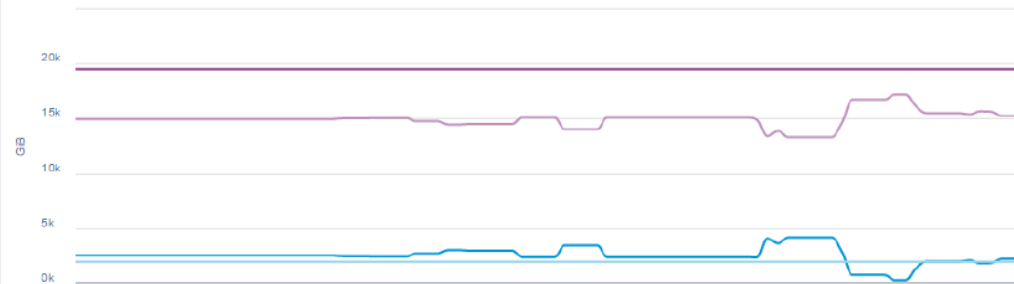
- This provides in depth information about the Chunklet utilization from the various drives in a system
- Most customers will be happy with the System Capacity report, this just provides more insight around what kind of Chunklet type is allocating space

✓ Physical Drive Space | Charts ▾

Summary

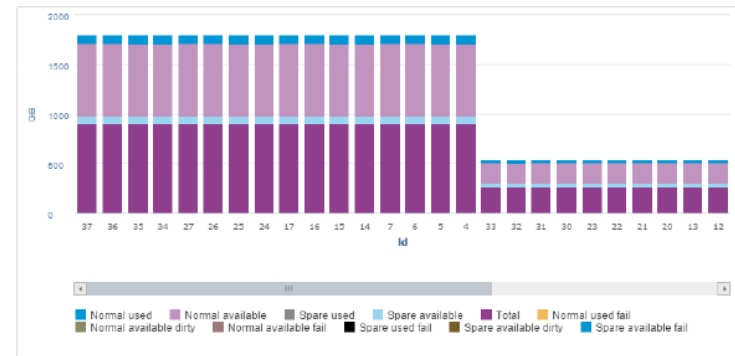
System	mktg-eos62	Report type	Sampling	Time interval	Daily	Metric types	Chunklet usage
Component	Physical drives						

Chunklet Usage



AT Time Report - Physical Drive Space

Chunklet Usage: 25-Apr-2015 00:00:00 o'clock PDT



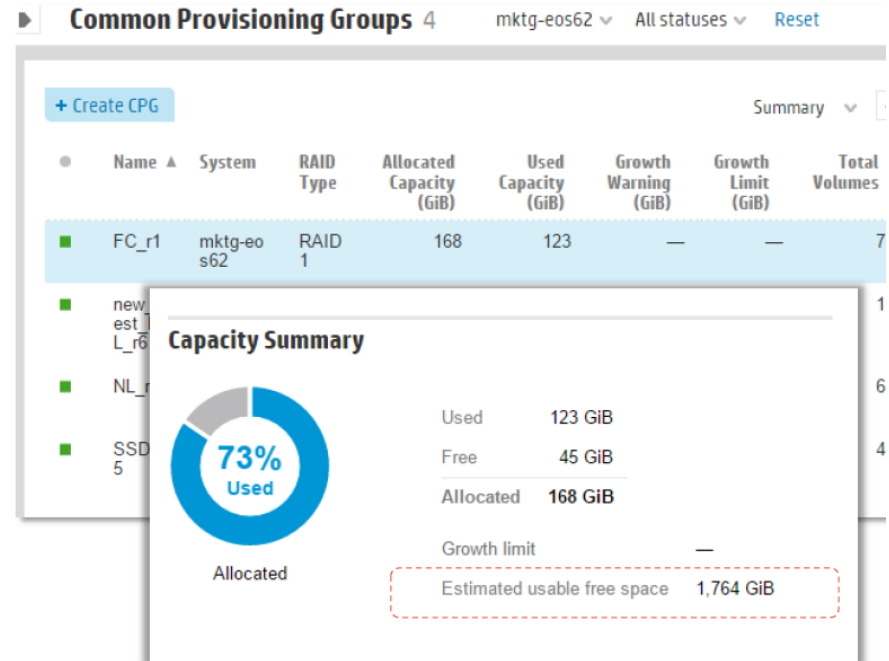
— Normal used
— Normal available
— Spare used
— Spare available
— Total

CPGs and Space Management

On a 3PAR system space is allocated on-demand from a CPG.

The amount of usable space that can be allocated can be a factor of the CPG characteristics and the available free physical space at any given point-in time.

- A CPG will grow based on its growth settings, growth will trigger creation or expansion of Logical Disk (LD) and the space consumed by the LDs will account as Allocated to the CPG.
- As new 16KB pages of writes are allocated to the LDs these will start consuming space and account as Used space.
- The Free space shows how many more pages can be allocated before the next CPG growth.
- The maximum amount of space a CPG can allocate is estimated and exposed to the user.





Performance & Troubleshooting



HPE 3PAR System Tuner

- ▶ HPE 3PAR System Tuner is an optional feature that improves performance by identifying over-used physical disks, and performing load balancing on those disks without interrupting access.
- ▶ The HPE 3PAR OS automatically creates a balanced system layout by mapping virtual volumes to many logical disks, and creating logical disks from chunklets drawn from many physical disks.
- ▶ The I/O for each volume is striped across many physical disks, increasing the throughput of the volume.



-
- ▶ As the system grows and new applications are introduced, new storage usage patterns can emerge, and the system performance can degrade.
 - ▶ System Tuner maintains peak system performance by automatically detecting and resolving bottlenecks without interrupting access.



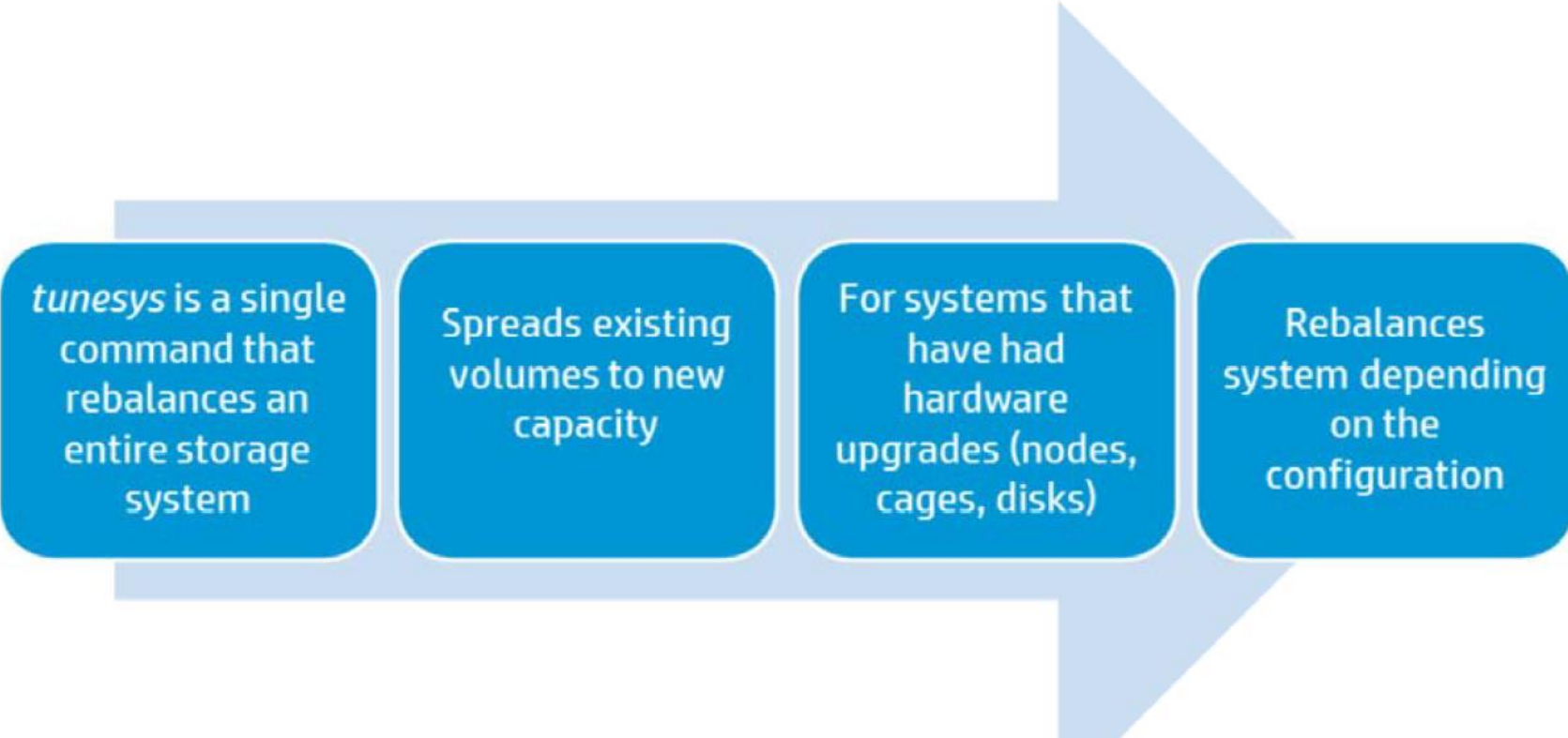
-
- ▶ There are two general reasons why a physical disk may have degraded performance:
 - ▶ The physical disk has reached its maximum throughput due to an unbalanced load. A disk in this state typically has unusually high average service times when compared to other disks.
 - ▶ The physical disk is a bad disk. A bad disk typically has unusually high maximum service times when compared to other disks.



-
- ▶ System Tuner allows you to:
 - ▶ Perform physical disk performance tuning on an entire system or on a specified subset of disks.
 - ▶ Set performance thresholds for physical disk tuning.
 - ▶ Identify and relocate under-performing chunklets.



Introduction to tunesys



tunesys is a single command that rebalances an entire storage system

Spreads existing volumes to new capacity

For systems that have had hardware upgrades (nodes, cages, disks)

Rebalances system depending on the configuration

Understanding performance

Common terms and definitions

Input/output operations per second (IOPS)

- Each component has a max IOPS value based on physical hardware and design
- Can vary greatly based on type of workload (such as pattern, RAID, RW ratio, or queue)

Bandwidth (bits/s, KB/s, MB/s, GB/s)

- Each component has a max bandwidth value based on physical hardware and design
- Can vary greatly based on type of workload (such as pattern, RAID, RW ratio, or queue)
- Based on the following equation and known variables: $\text{Bandwidth} = \text{IOPS} \times (\text{block size of IO})$

Service time

- Time to process one request (I/O) once received
- In some cases, this might include any wait time as well
- Can vary greatly based on type of workload

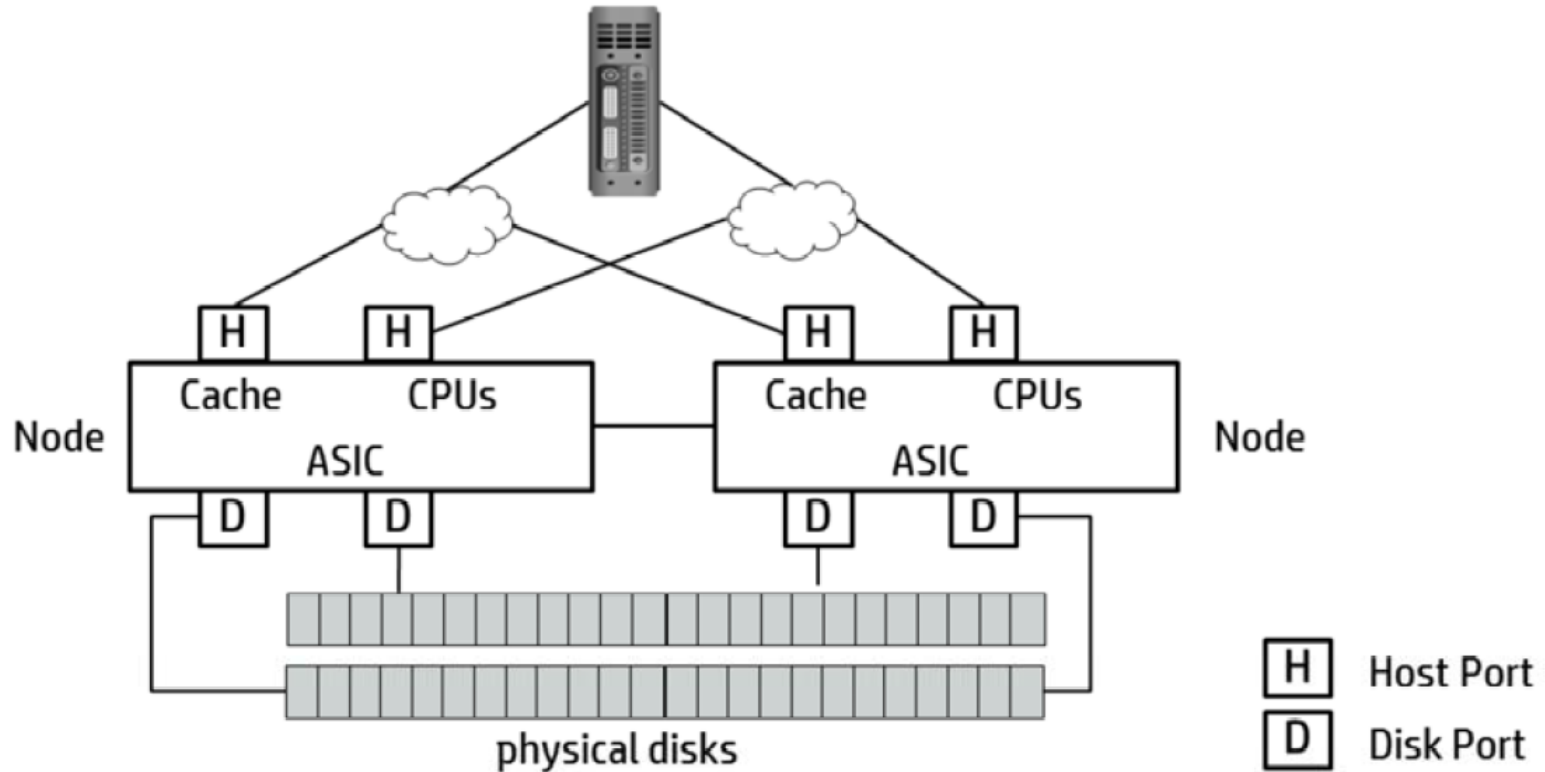


Servicing I/O—Array basics

- All arrays have:
 - Host ports for server connectivity
 - CPU and cache to process commands and data
 - Internal buses for cache coherency
 - Back-end buses connecting CPUs to physical disks
 - RAID groupings to provide redundancy for the data
 - Physicals disks to store data
- Performance problems happen when one or more of these components is stressed
 - Simple rule of thumb 1—When IOPS or bandwidth are exceeded, we will see higher service times
- Performance troubleshooting is to identify which components are stressed and provide alternatives in configuration or workload



3PAR component block diagram



Hardware component limits—CPU

CPU limits

CPU can limit your performance if maxed out

- Causes 3PAR to not be able to reach max performance
- Even if only one node is at max, can cause the entire system to run slow



Which processes take a lot of CPU?

- Active RC nodes have higher CPU utilization (30 to 50% higher)
- SSH connections use more CPU than CLI sessions
- Very high I/O rates (more read-heavy loads)



Hardware component limits—Cache

Delayed ack mode

“Delayed ack” is a behavior of HP 3PAR systems:

- When the cache is filled faster than it can be destaged to disk
- Most frequently caused by physical disk performance limitations
- Determined by the number of dirty cache pages for a type of disk exceeding 85% of the allowed maximum

Symptoms of delayed ack include:

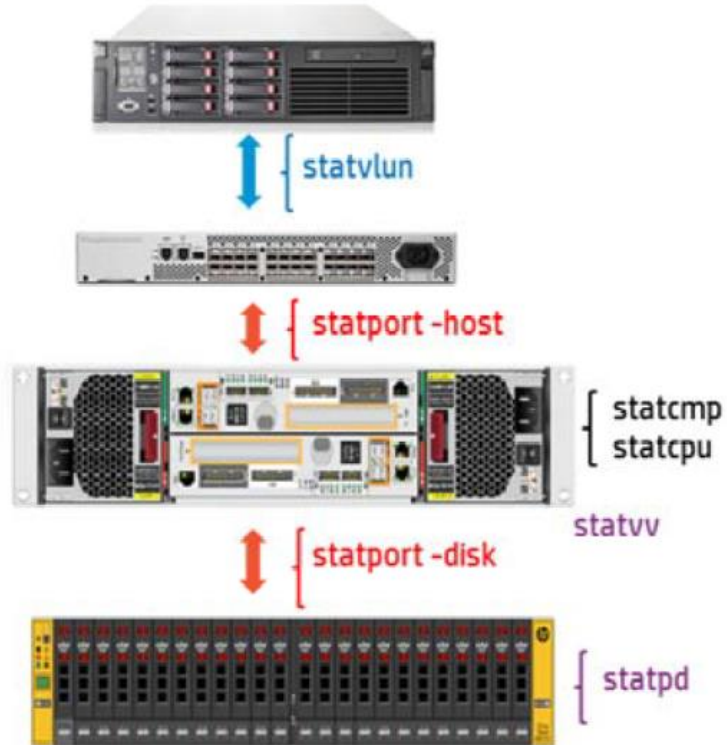
- The system starts delaying incoming write I/Os to keep the number of dirty pages below 85%
- Increasing the number of outstanding I/Os only results in a higher average response time



CLI stat commands

Six layers to check

- Host VLUNs
- Host ports
- Node cache/CPU
- Node volumes
- Disk ports
- Disks



dx-3par cli% showalert -n

Id : 30
State : New
Message Code: 0x00400fa
Time : 2020-07-04
Severity : Major
Type : Component
Message : Virtual Volume
Failed (Internal Consistency Error)

dx-3par cli% showbattery -d

-----Node 0 PS 0 Battery 0-----

Node ID	:	0
Power Supply ID	:	0
Battery ID	:	0
Manufacturer	:	XYRATEX
Assembly Part	:	--
Assembly Serial Number	:	--
State	:	Failed
Detail State	:	Failed
Charge Level(%)	:	0
Expired	:	No
Test in Progress	:	No
Charge State	:	--
Max Battery Life(Whrs)	:	16
Expiration Date	:	n/a

dxs-3par cli% showcage

Id	Name	LoopA	Pos.A	LoopB	Pos.B	Drives	Temp	RevA	RevB	Model	FormFactor
0	cage0	0:2:1	0	1:2:1	0	24	20-20		4044	DCS8	SFF
1	cage1	0:4:1	0	1:4:1	0	24	20-20		4044	DCS8	SFF
2	cage2	0:0:1	0	1:1:1	0	24	20-20			DCN2	SFF
3	cage3	0:1:1	0	1:0:1	0	24	20-20	-		DCS7	LFF

dxs-3par cli% showcage -i

-----Midplane-----

Cage	-Name-	-Manufacturer-	-Assem_Part-	-Assem_Serial-	-Firmware-	-Type-	-----W
0	cage0	XYRATEX	n/a	n/a	--	DCS8	50050CC
30AF874F0							
1	cage1	XYRATEX	n/a	n/a	--	DCS8	50050CC
30AF974F0							
2	cage2	XYRATEX	n/a	n/a	--	DCN2	50050CC
30AF774F0							
3	cage3	XYRATEX	n/a	n/a	--	DCS7	50050CC
30AFA74F1							

-----Interface Card-----

Cage	-I/F_card-	-Name-	-Manufacturer-	-Assem_Part-	-Assem_Serial-	-Firmware-
0	0	SAS A	XYRATEX	n/a	n/a	

dxs-3par cli% statcache -iter 1

```
00:29:38 07/08/2020 ----- Current -----
```

			CMP	FMP	Total				CMP	FMP	Total
Node	Type	Accesses	Hit%	Hit%	Hit%	Accesses	Hit%	Hit%	Hit%		
0	Read	0	0	0	0	0	0	0	0	0	0
0	Write	0	0	0	0	0	0	0	0	0	0
1	Read	0	0	0	0	0	0	0	0	0	0
1	Write	0	0	0	0	0	0	0	0	0	0

Internal Flashcache Activity

```
----- Current -----
```

Node	Type	Accesses	IO/s	MB/s	Accesses	IO/s	MB/s
0	Read Back	0	0	0	0	0	0
0	Destaged Write	0	0	0	0	0	0
1	Read Back	0	0	0	0	0	0
1	Destaged Write	0	0	0	0	0	0

```
----- Total -----
```

----- FMP Oueue Statistics -----

dx-c-3par cli% statcache -v

```
00:30:44 07/08/2020      ----- Current -----      ----- Total -----
                                CMP  FMP Total                CMP  FMP Total
    VVid VVname      Type Accesses Hit% Hit%  Hit%      Accesses Hit% Hit%  Hit%

- Internal Flashcache Activity -
- Current --              -- Total ---
VVid VVname Type      Accesses              Accesses
```

Press the enter key to stop...

```
00:30:46 07/08/2020      ----- Current -----      ----- Total -----
                                CMP  FMP Total                CMP  FMP Total
    VVid VVname      Type Accesses Hit% Hit%  Hit%      Accesses Hit% Hit%  Hit%

- Internal Flashcache Activity -
- Current --              -- Total ---
VVid VVname Type      Accesses              Accesses
```

dxs-3par cli% statcpu

00:32:50 07/08/2020

node,cpu	user	sys	idle	intr/s	ctxt/s
0,0	3	80	18		
0,1	2	51	48		
0,2	1	62	37		
0,3	0	88	12		
0,total	2	70	29	6971	9102
1,0	6	82	12		
1,1	5	77	18		
1,2	3	83	15		
1,3	2	57	42		
1,total	4	75	22	6770	8383

Press the enter key to stop...



dxs-3par cli% **statiscsi**

00:33:47 07/08/2020 -----From start of statiscsi command-----

-----Receive----- ---Transmit--- -----Total-----

port	Protocol	Pkts/s	KBytes/s	Pkts/s	KBytes/s	Pkts/s	KBytes/s	Errs/s
Total	Eth	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total	IP	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total	TCP	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total	iSCSI	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Press the enter key to stop...

dxs-3par cli% statport

00:34:29 07/08/2020		r/w	I/O per second			KBytes per sec			Svt ms		IOSz KB		
Port	D/C		Cur	Avg	Max	Cur	Avg	Max	Cur	Avg	Cur	Avg	Qlen
0:0:1	Data	t	216	216	216	7085	7085	7085	7.58	7.58	32.8	32.8	2
0:1:1	Data	t	200	200	200	6551	6551	6551	6.64	6.64	32.8	32.8	3
0:2:1	Data	t	191	191	191	6274	6274	6274	5.60	5.60	32.8	32.8	1
0:4:1	Data	t	204	204	204	6668	6668	6668	7.06	7.06	32.8	32.8	1
0:9:1	Data	t	0	0	0	0	0	0	0.00	0.00	0.0	0.0	0
1:0:1	Data	t	224	224	224	7350	7350	7350	6.25	6.25	32.8	32.8	3
1:1:1	Data	t	240	240	240	7853	7853	7853	6.40	6.40	32.8	32.8	0
1:2:1	Data	t	209	209	209	6848	6848	6848	5.75	5.75	32.8	32.8	0
1:4:1	Data	t	220	220	220	7212	7212	7212	6.50	6.50	32.8	32.8	0
1:9:1	Data	t	0	0	0	0	0	0	0.00	0.00	0.0	0.0	0

10	Data	t	1704	1704		55841	55841		6.48	6.48	32.8	32.8	10

Press the enter key to stop...

dxs-3par cli% statvln

		I/O per second			KBytes per sec			Svt ms		IOSz KB		
Lun	VVname	Host	Port	Cur	Avg	Max	Cur	Avg	Max	Cur	Avg	Qlen
0	t			0	0		0	0		0.00	0.00	0.0 0.0 0

Press the enter key to stop...

		I/O per second			KBytes per sec			Svt ms		IOSz KB		
Lun	VVname	Host	Port	Cur	Avg	Max	Cur	Avg	Max	Cur	Avg	Qlen
0	t			0	0		0	0		0.00	0.00	0.0 0.0 0

Press the enter key to stop...



dxs-3par cli% **checkhealth**

Checking alert

Checking ao

Checking cabling

Could not check cabling: Error: Unable to get topology on port 1:4:1

Checking cage

Checking cert

Checking dar Checking vlun

Checking vv

Checking sp

Component	Summary Description	Qty
Alert	New alerts	39
Component	Could not check cabling	1
Cage	Cages not on current firmware	4
Cage	Degraded or failed cage power supply fans	1
Cage	Degraded or failed cage power supplies	1
File	Behavior altering files	10
File	Nodes with Dump or HBA core files	2
File	sr_mnt not mounted	1
File	Admin Volume is not mounted	1
Node	Nodes with non-zero BIOS skip mask	2
Node	Power supplies with failed or degraded batteries	2
Component	Could not check pd	1
QoS	Unable to check QoS	1
vlun	Hosts not connected to a port	3
VV	Degraded or failed virtual volumes	18
VV	Expired volumes	1

16 total

88

```
dxs-3par cli% checkport -iter 1 0:4:1
```

```
Error: Port 0:4:1 shows 26 devices present, loopback not allowed in this case.
```

```
dxs-3par cli% checkport -iter 1 1:9:1
```

```
Error: Port 1:9:1 state must be 'ready', not 'config_wait'
```



```
Specified volumes esx2 will be unavailable during this operation.
Do you want to continue with this operation?
```

```
Task 123 has been started to repair administration information for VVs: esx2
```

```
dxcc-3par cli% showvv esx*
```

Lab



Thank You ...

